APPROACH TO STROKE

CHAIRPERSON- DR PRABHU

PRESENTER- DR SAICHARAN

WHAT IS STROKE?

- A stroke, or cerebrovascular accident, is defined as an abrupt onset of a neurological deficit that is attributable to a focal vascular cause.

Harrison's principles of Internal Medicine, 20th edition

Stroke



- Overall second most common cause of death
- Stroke continues to be a major public health problem
- more often disabling than fatal, stroke is the leading cause of severe neurologic disability and results in enormous costs.

Prevalence—2-5/1000 population
Incidence---0.4-2/1000 Population
30-50 patients/day in Bangalore

GBD2011 and WHO 2016

CASE SCENARIO

- 57 year old male presented with h/o weakness in the rt side of the body and deviation of angle of mouth to the right side after waking up in the morning.
- h/o episode of such weakness in rt side 1 month back lasting for about 6 hours which recovered without any residual deficit

CASE SCENARIO

- K/C/O Type2 diabetes poorly controlled since last 15 years
- k/c/o Hypertension since 10 years on treatment

- Diagnosis?

CASE SCENARIO

- CT scan shows Acute infarct involving Lt. Internal capsule and chronic small vessel ischemic changes

- What Next?

Transient Ischaemic Attack (TIA)

- An acute focal neurological deficit resulting from cerebrovascular disease with resolution of signs and symptoms within 24 hours.
- Tissue-based definition of TIA: a transient episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia, without acute infarction. [ASA]

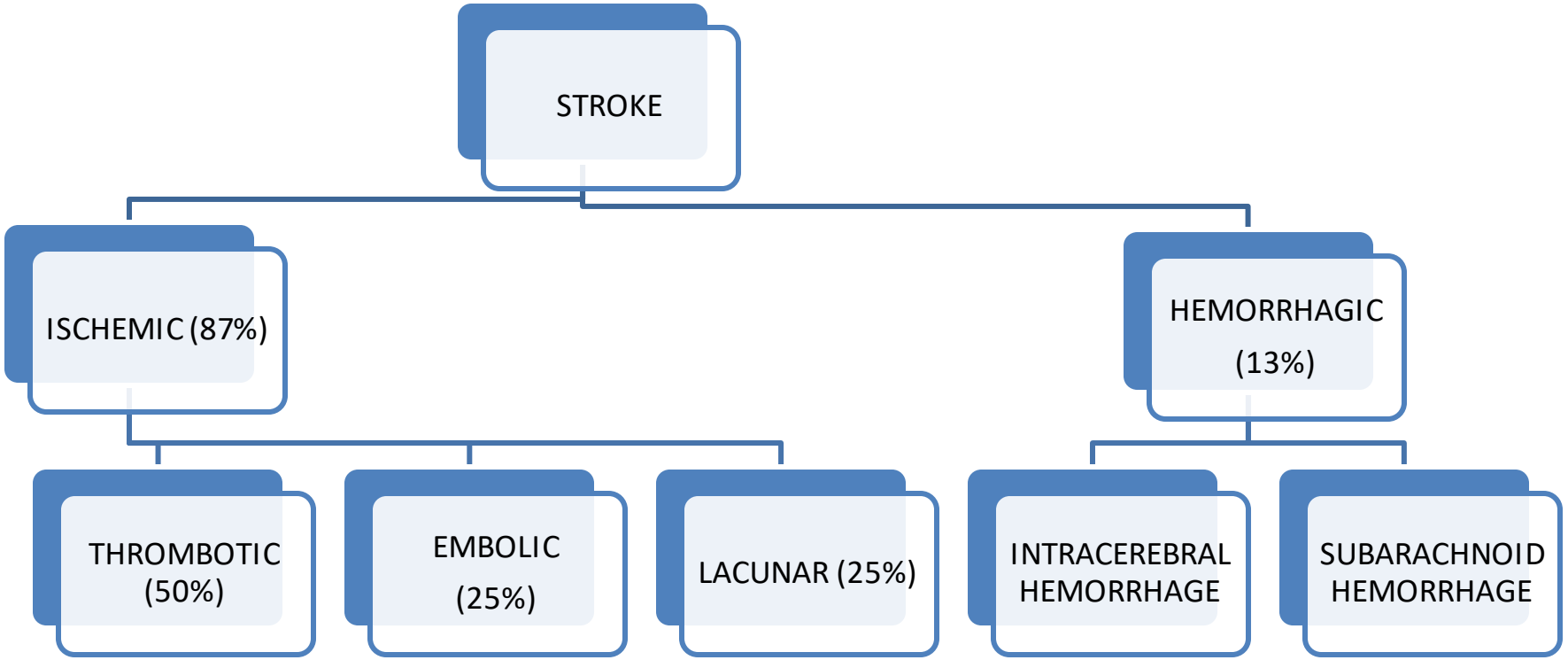
- Reversible Ischaemic Neurological Deficit (RIND) – attack lasting longer than 24 hours but with complete clearance of signs and symptoms within 3 weeks.

- Completed Stroke – neurological deficit lasts longer than 7 days.

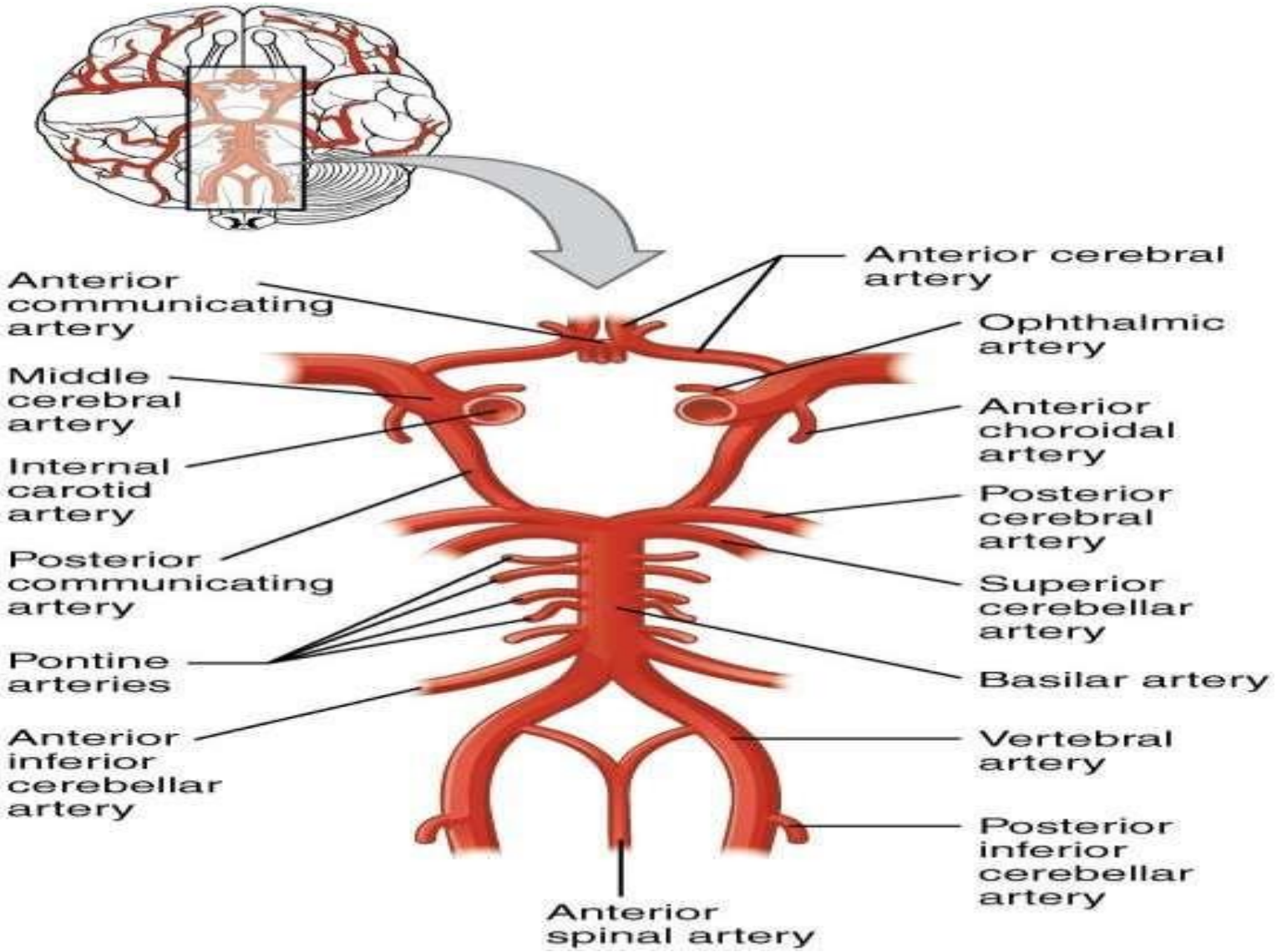
STROKE MIMICS

- Seizure
- Intracranial tumor
- Migraine
- Metabolic encephalopathy
- Hypoglycemia
- Syncope

WHO STROKE CLASSIFICATION



BLOOD SUPPLY OF BRAIN

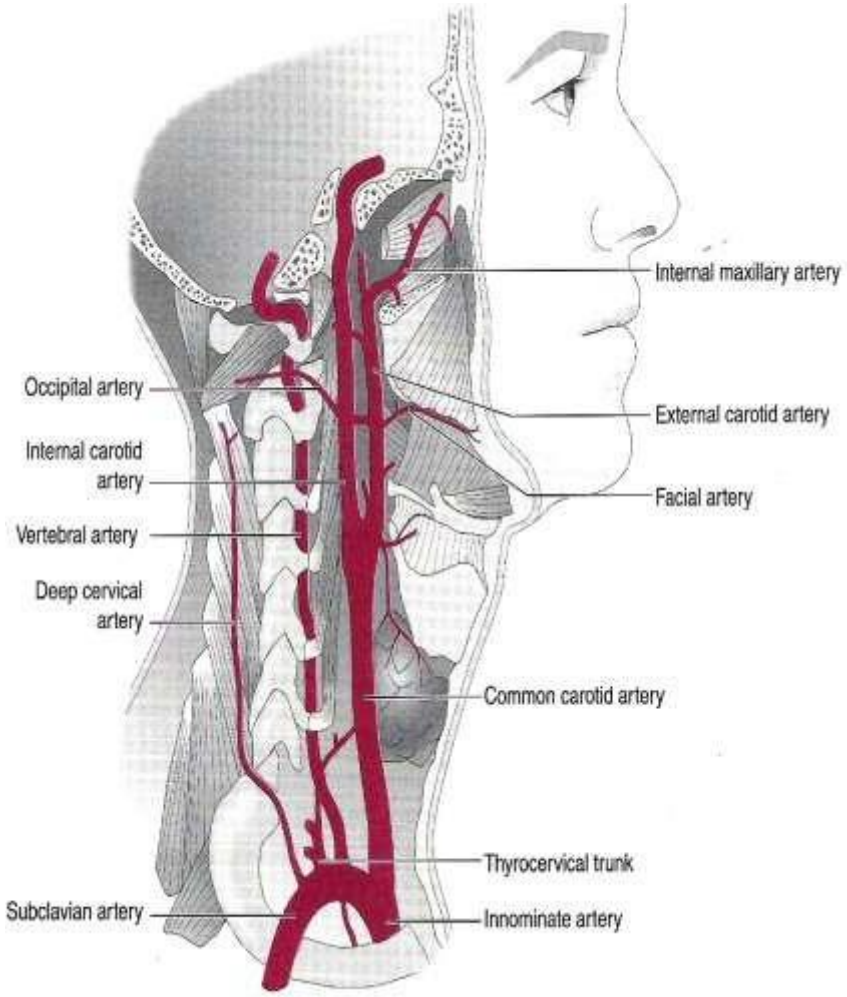


CAROTID ARTERY SYNDROMES

- COMMON CAROTID OCCLUSION (~1%)
- PATHOLOGY:
  - Atheromatous plaque m/c origin of lt. common carotid
  - Radiation therapy to neck
- Mostly asymptomatic as retrograde flow compensates

- INTERNAL CAROTID OCCLUSION (99%)
- PATHOLOGY:
  - Athero-thrombotic narrowing
  - Dissection
  - Vasculitis (rarely)
- Territory involved depends on level of occlusion

Internal carotid artery infarction

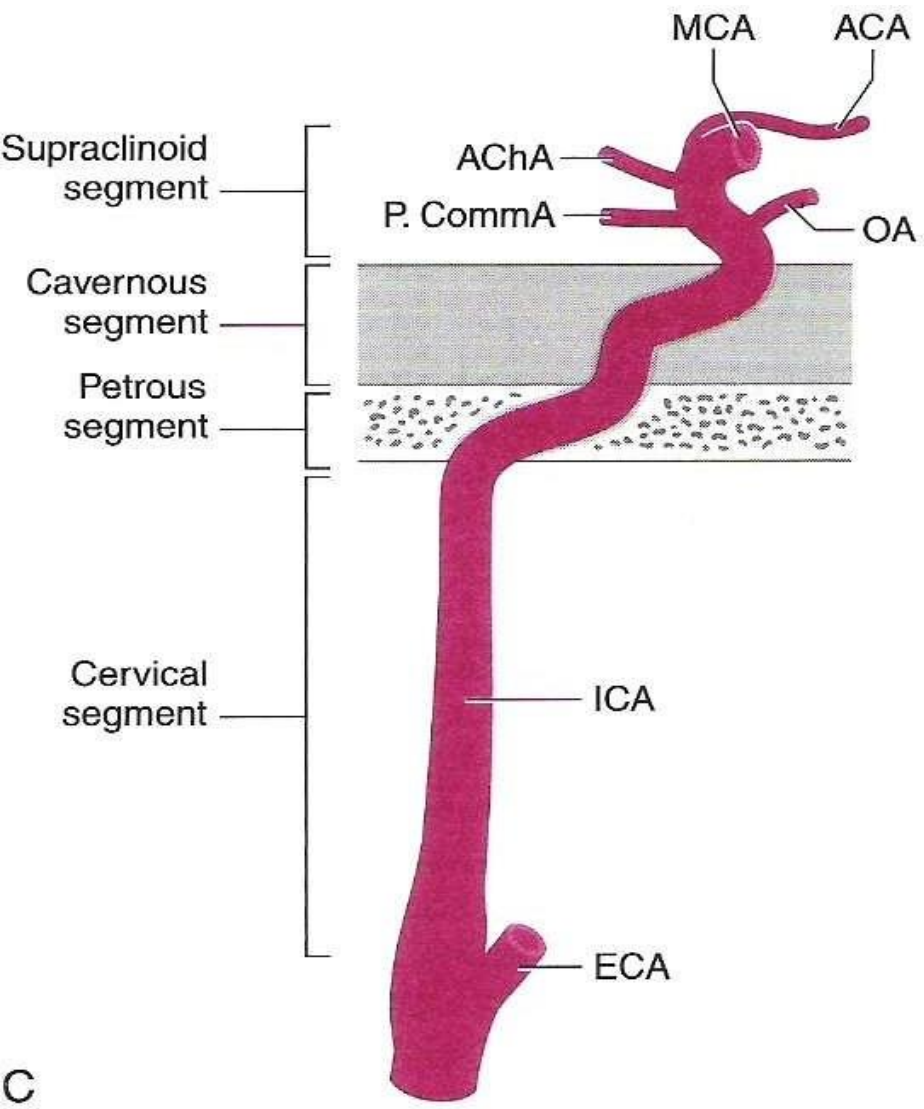


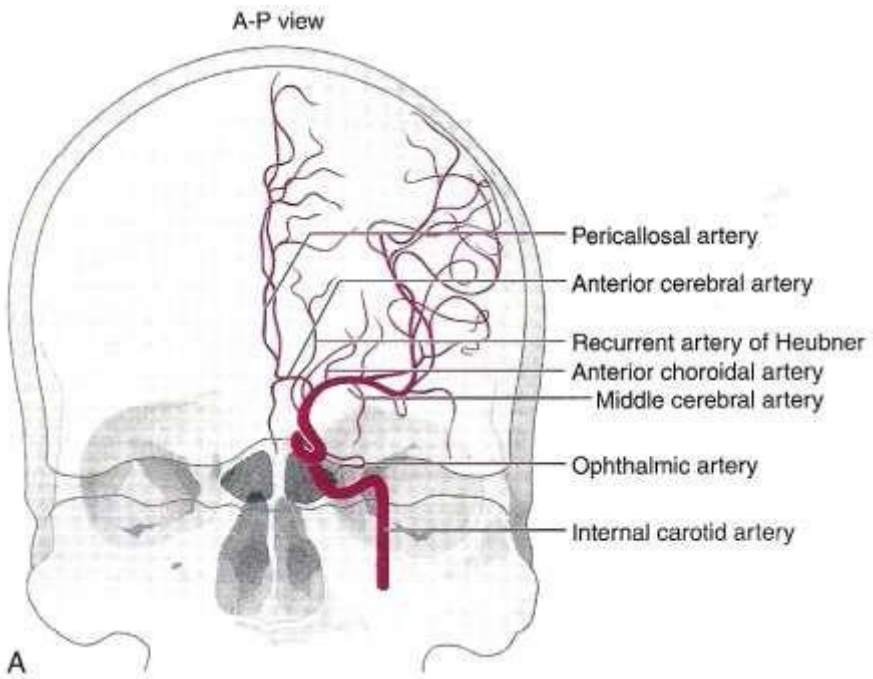
Clinical features

Variable - based on the collaterals and mechanism of stroke (embolism, extension of thrombus , low flow)

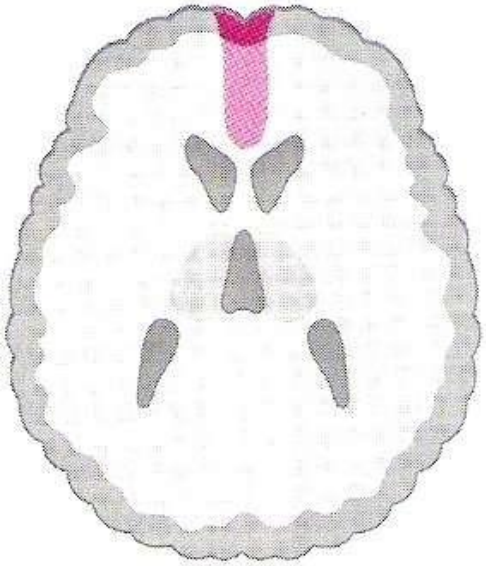
1. Amaurosis fugax
2. Watershed infarctions
3. MCA/ACA- either alone or in combinations

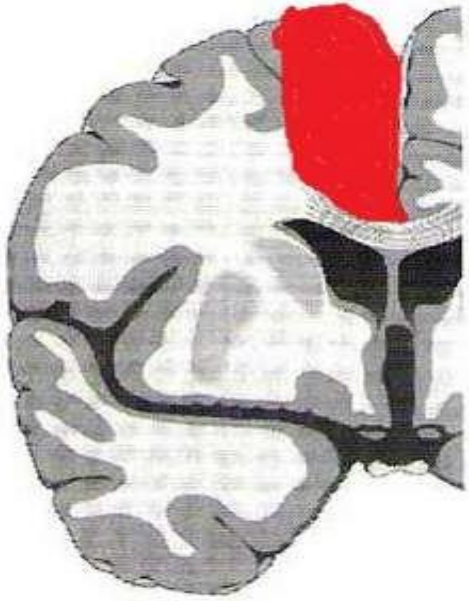
STROKE WITHIN ANTERIOR CIRCULATION





Anterior cerebral artery infarction





Clinical features

1. Contralateral
   a. paralysis of leg and foot with paresis of arm
   b. cortical sensory loss over leg and foot
   c. presence of primitive reflexes
2. Urinary incontinence
3. Gait apraxia
4. Mutism, delay and lack of spontaneity of motor acts
5. Apraxia of left sided limbs(with left sided lesion and corpus callosum involvement)

STROKE WITHIN ANTERIOR CIRCULATION

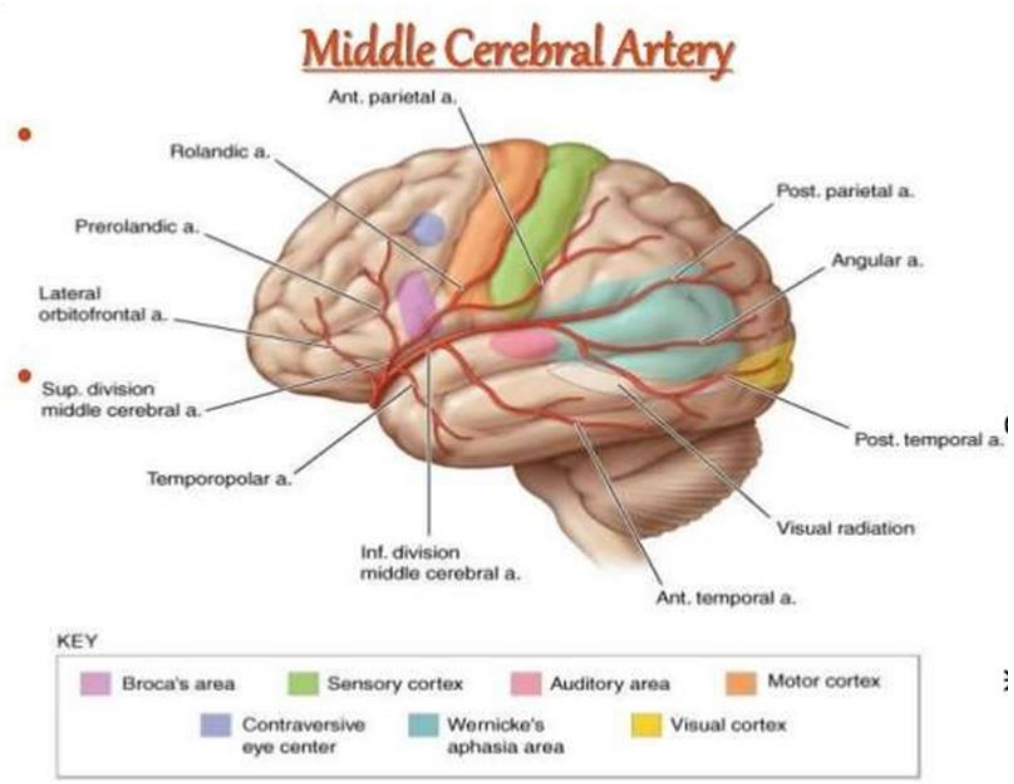
- ANTERIOR CHOROIDAL ARTERY:

- Supplies posterior limb of internal capsule, retrolentiform and sublentiform parts

- Syndrome comprises
  - c/l hemiplegia
  - c/l hemianaesthesia
  - c/l homonymous hemianopia

STROKE WITHIN ANTERIOR CIRCULATION

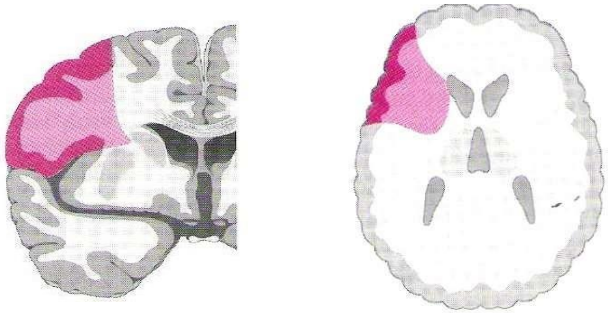
- MIDDLE CEREBRAL ARTERY: (m/c embolic)



MIDDLE CEREBRAL ARTERY INFARCTION

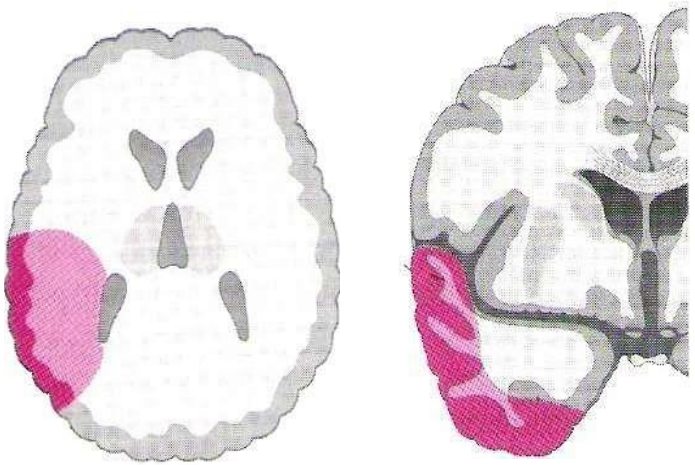
SUPERIOR BRANCH

- Contralateral hemiplegia – face and upper limb more involved than lower limb.
- Contralateral hemisensory loss.
- Conjugate gaze paresis(patient looks
- Broca's dysphasia (if left sided)



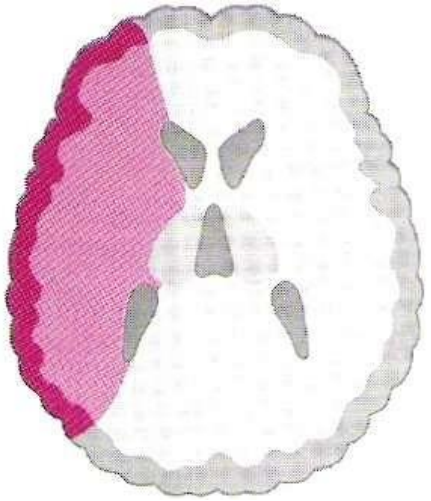
INFERIOR BRANCH

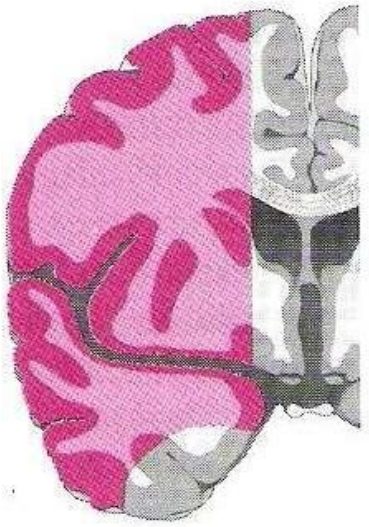
- Contralateral hemianopia.
- Wernicke's dysphasia (if left sided)
- Left spatial neglect (if right sided)



MIDDLE CEREBRAL ARTERY INFARCTION - STEM OCCLUSION

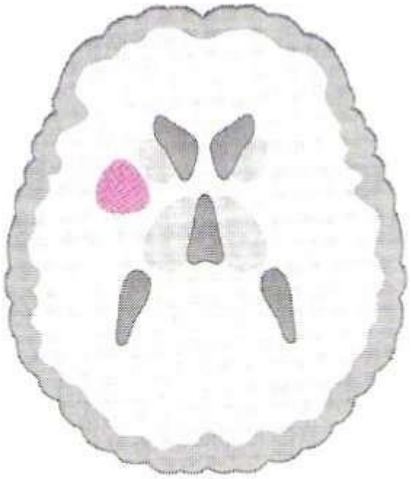
- Clinical features
  - Contralateral hemiplegia
  - Contralateral hemisensory loss
  - Contralateral gaze palsy
  - Contralateral hemianopia
  - Global dysphasia (Left sided lesion)
  - Anosognosia and amorphosynthesis (Right sided lesion)
  - Altered sensorium (due to edema)

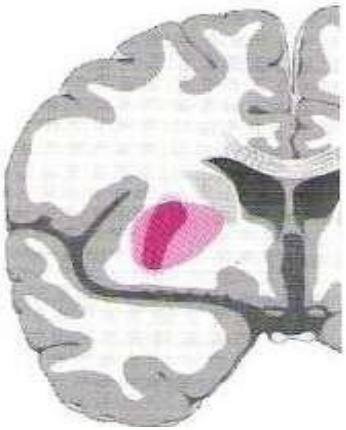




MIDDLE CEREBRAL ARTERY INFARCTION- LENTICULOSTRIATAL OCCLUSION

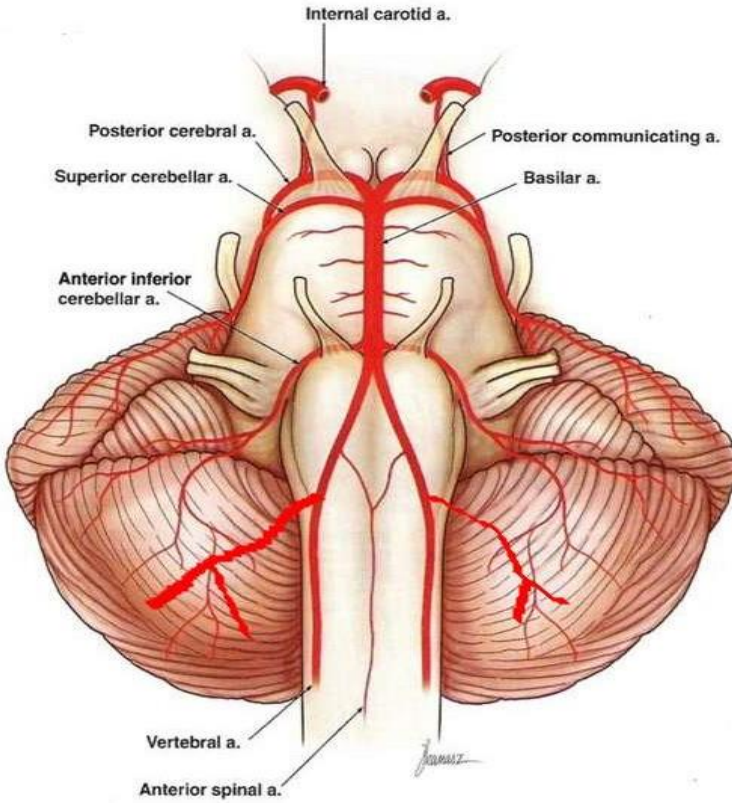
- Deep penetrating or lenticulostriate branches– Internal capsule, caudate nuclues, putamen and outer pallidus
- Occlusion of lenticulostriate branches-
  - If ischemia of internal capsule produces pure motor or sensorymotor stroke contralateral to the side of lesion
  - If ischemia of putamen, pallidus- predominantly parkinsonian features



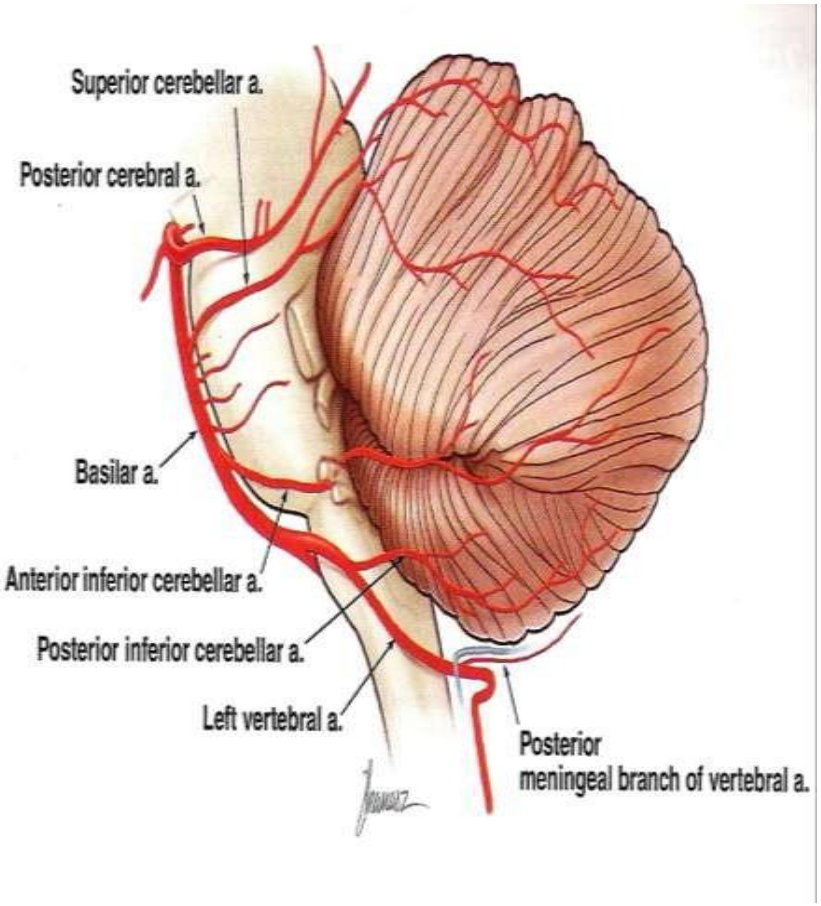


STROKE WITHIN POSTERIOR CIRCULATION

- POSTERIOR CEREBRAL ARTERY: (M/C: ATHEROMA, EMBOLI)



ventral view



STROKE WITHIN POSTERIOR CIRCULATION

- P1 SYNDROMES:

  Midbrain, subthalamic, and thalamic signs, which are due to disease of the proximal P1 segment of the PCA or its penetrating branches (thalamogeniculate, Percheron, and posterior choroidal arteries)

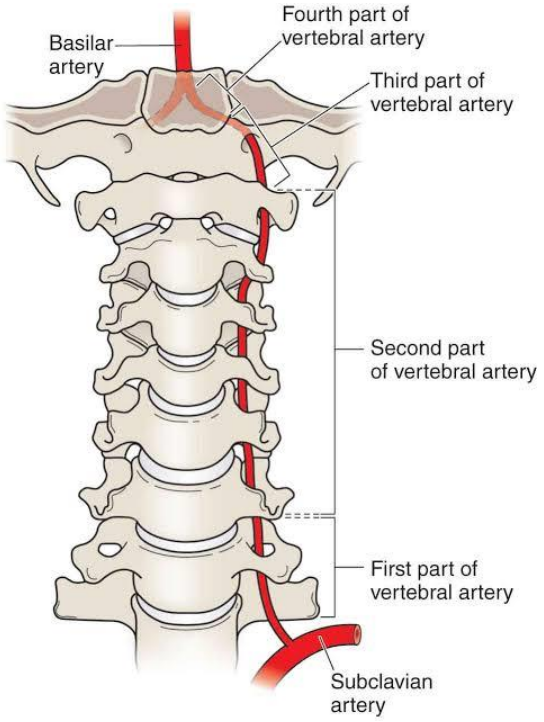
STROKE WITHIN POSTERIOR CIRCULATION

- P2 SYNDROMES:

  Cortical temporal and occipital lobe signs, due to occlusion of the P2 segment distal to the junction of the PCA with the posterior communicating artery.

STROKE WITHIN POSTERIOR CIRCULATION

- VERTEBRAL AND PICA SYNDROMES:
  - Atherothrombotic lesions have a predilection for V1 and V4 segments of the vertebral artery.
  - The first segment may become diseased at the origin of the vessel and may produce posterior circulation emboli and Embolic occlusion or thrombosis of a V4 segment causes ischemia of lateral medulla.
  - the second and third segments of the vertebral artery is affected more commonly by dissection, fibromuscular dysplasia, and, rarely, encroachme by osteophytic spurs within the vertebral forami and atherosclerotic lesions are rarely seen.



STROKE WITHIN POSTERIOR CIRCULATION

- BASILAR ARTERY:

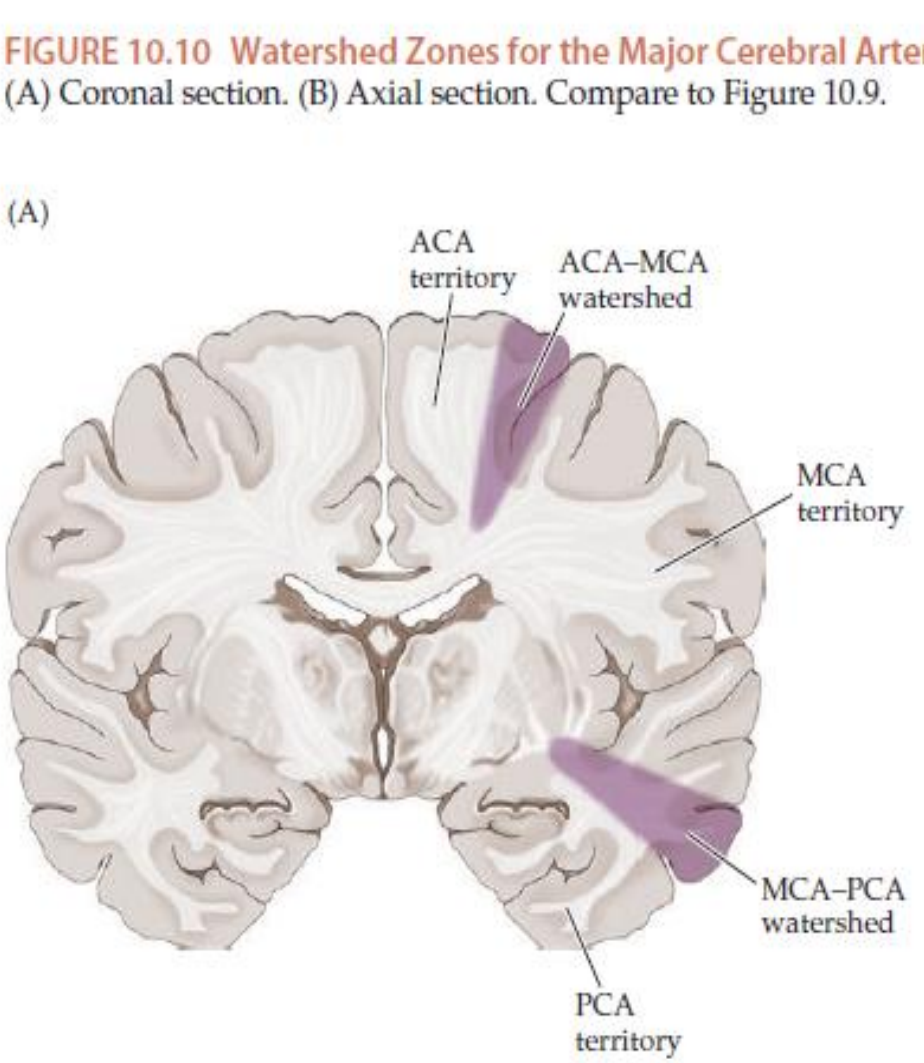
- Occlusion of basilar artery-b/l brainstem signs.
- Occlusion of basilar branch artery- unilateral motor, sensory and cranial nerves.
- Complete basilar artery occlusion(Locked in state)- b/l long tract(sensory/motor) with cranial nerve and cerebellar dysfunction- preserved consciousness, quadriplegia and cranial nerve signs.

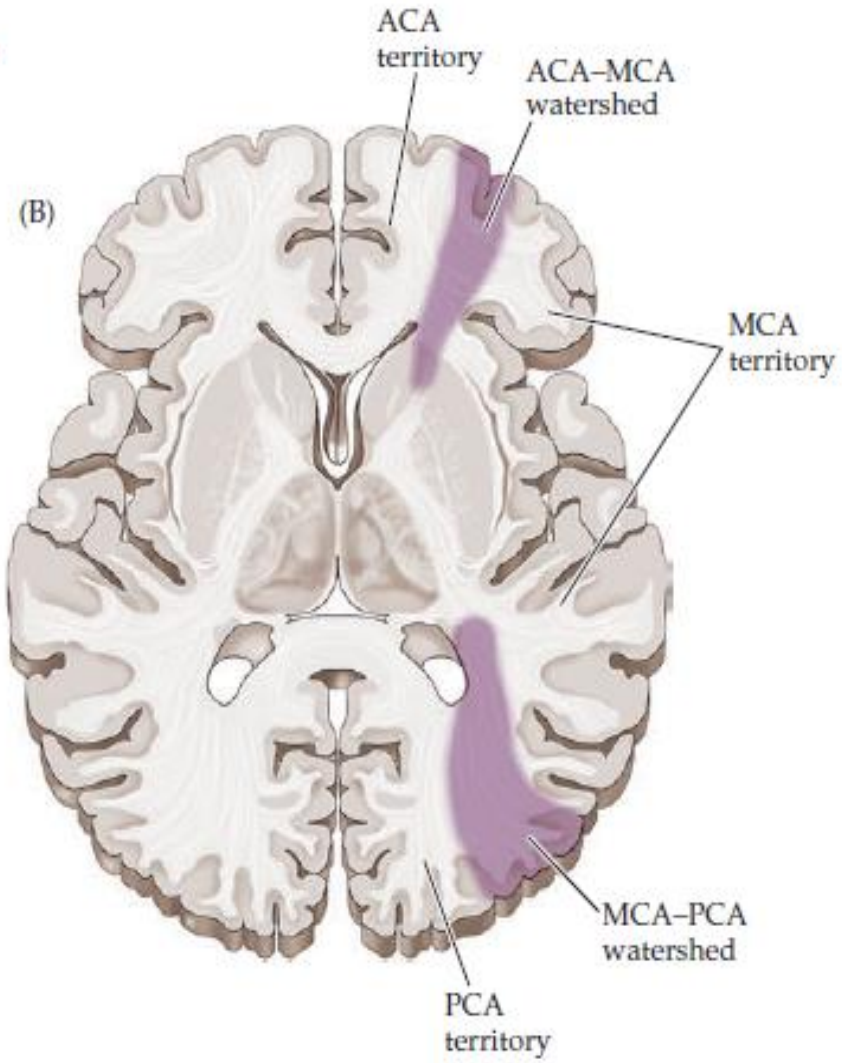
WATERSHED INFARCTS

- When the blood supply to two adjacent cerebral arteries is compromised, the regions between the two vessels are most susceptible to ischemia and infarction. These regions between cerebral arteries are called **watershed zones**
- **Bilateral watershed infarcts in both the** ACA–MCA and MCA–PCA watershed zones can occur with severe drops in systemic blood pressure.
- Man in barrel syndrome

WATERSHED INFARCTS

FIGURE 10.10 Watershed Zones for the Major Cerebral Arteries
(A) Coronal section. (B) Axial section. Compare to Figure 10.9.





SMALL-VESSEL STROKE

- The term lacunar infarction refers to infarction following atherothrombotic or lipohyalinotic occlusion of a small artery in the brain. The term small-vessel stroke denotes occlusion of such a small penetrating artery and is now the preferred term.
- Small-vessel strokes account for ~20% of all strokes.
- The MCA stem, the arteries comprising the circle of Willis and the basilar and vertebral arteries all give rise to 30- to 300-µm branches that penetrate the deep gray and white matter of the cerebrum or brainstem
- Thrombosis of these vessels causes small infarcts that are referred to as lacunes (Latin for “lake” of fluid noted at autopsy). These infarcts range in size from 3 mm to 2 cm in diameter.

SMALL-VESSEL STROKE

- Hypertension and age are the principal risk factors.
- The most common small-vessel stroke syndromes are the following:
  - Pure motor hemiparesis from an infarct in the posterior limb of the internal capsule or the pons; the face, arm, and leg are almost always involved;
  - pure sensory stroke from an infarct in the ventral thalamus;
  - ataxic hemiparesis from an infarct in the ventral pons or internal capsule;
  - dysarthria and a clumsy hand or arm due to infarction in the ventral pons or in the genu of the internal capsule.

MANAGEMENT

TRANSIENT ISCHEMIC ATTACK

ABCD2 SCORING

Patients with TIA score points (indicated in parentheses) for each of the following factors:

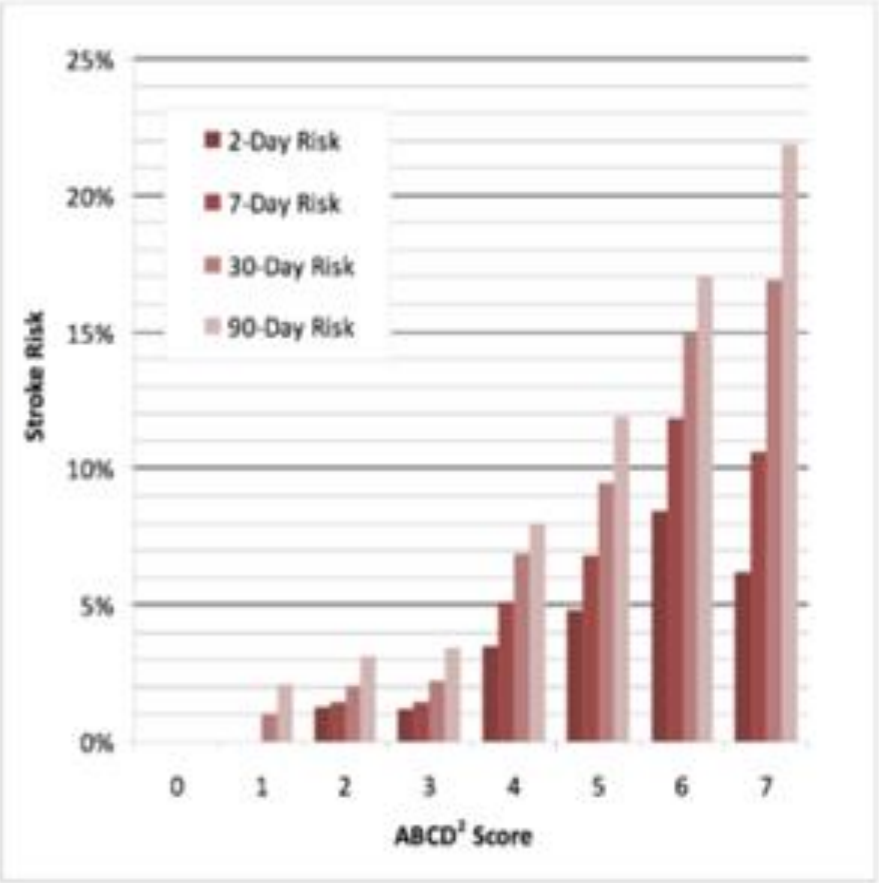
Age ≥60 years (1)
Blood pressure ≥140/90 mmHg on first evaluation (1)
Clinical symptoms of focal weakness with the spell (2) or speech impairment without weakness (1)
Duration ≥60 minutes (2) or 10 to 59 minutes (1)
Diabetes (1)

ABCD2 SCORING

Using the ABCD^2 Score

Higher ABCD^2 scores are associated with greater risk of stroke during the 2, 7, 30, and 90 days after a TIA (Figure). The authors of the ABCD^2 score made the following recommendations for hospital observation:[1]

ABCD^2 Score	2-day Stroke Risk	Comment
0-3	1.0%	Hospital observation may be unnecessary without another indication (e.g., new atrial fibrillation)
4-5	4.1%	Hospital observation justified in most situations
6-7	8.1%	Hospital observation worthwhile



[1] Johnston SC, Rothwell PM, Huynh-Huynh MN, Giles MF, Elkins JS, Sidney S, "Validation and refinement of scores to predict very early stroke risk after transient ischemic attack," *Lancet*, 369:283-292, 2007.

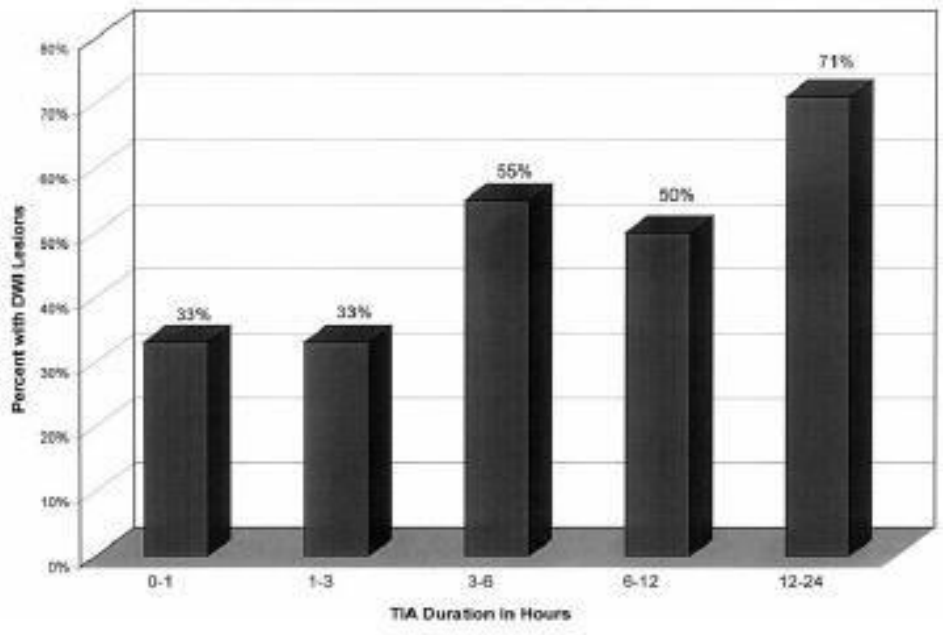
It is reasonable to hospitalize patients with TIA if they present within 72 hours of the event and any of the following criteria are present:

- ABCD ^{2} scores of 3 (**Class IIa, Level of Evidence C**).
- ABCD ^{2} scores of 0 to 2 and uncertainty that diagnostic workup can be completed within 2 days as an outpatient (**Class IIa, Level of Evidence C**).
- ABCD ^{2} scores of 0 to 2 and other evidence that indicates the patient's event was caused by focal ischemia (**Class IIa, Level of Evidence C**).

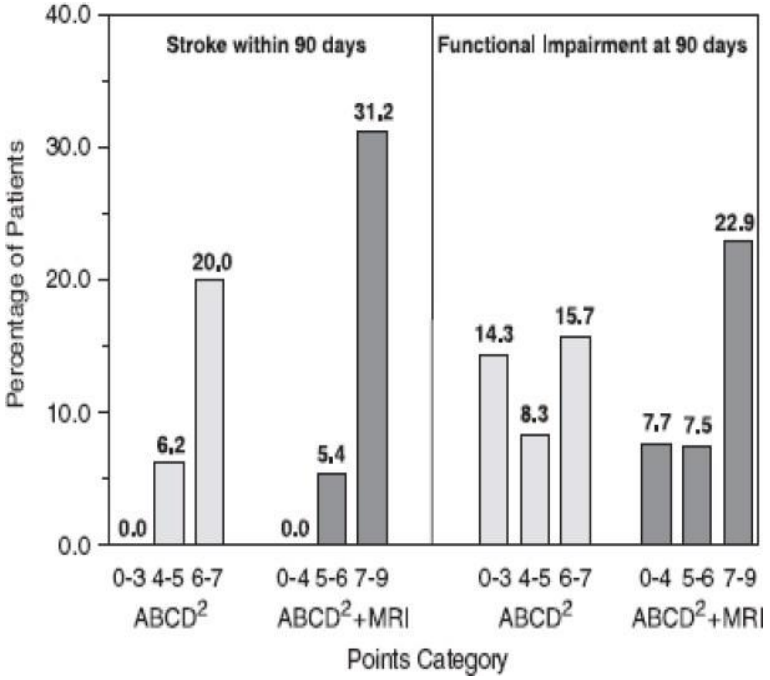
DIAGNOSTIC EVALUATION

- Patients with TIA should preferably undergo neuroimaging evaluation within 24 hours of symptom onset.
- **MRI**, including **DWI**, is the preferred brain diagnostic imaging modality.
- If MRI is not available, **CT** should be performed (**Class I, Level of Evidence B**).

Diffusion MRI in patients with TIA



ABCD2 +MRI (DWI)



Coutts et al. Int J. Stroke 2008; Ann Neurol 2005

NEUROVASCULAR EVALUATION

- Noninvasive imaging of the cervicocephalic vessels should be performed routinely as part of the evaluation of patients with suspected TIAs (**Class I, Level of Evidence A**).
- Noninvasive testing of the intracranial vasculature reliably excludes the presence of intracranial stenosis (**Class I, Level of Evidence A**) and is reasonable to obtain when knowledge of intracranial steno-occlusive disease will alter management

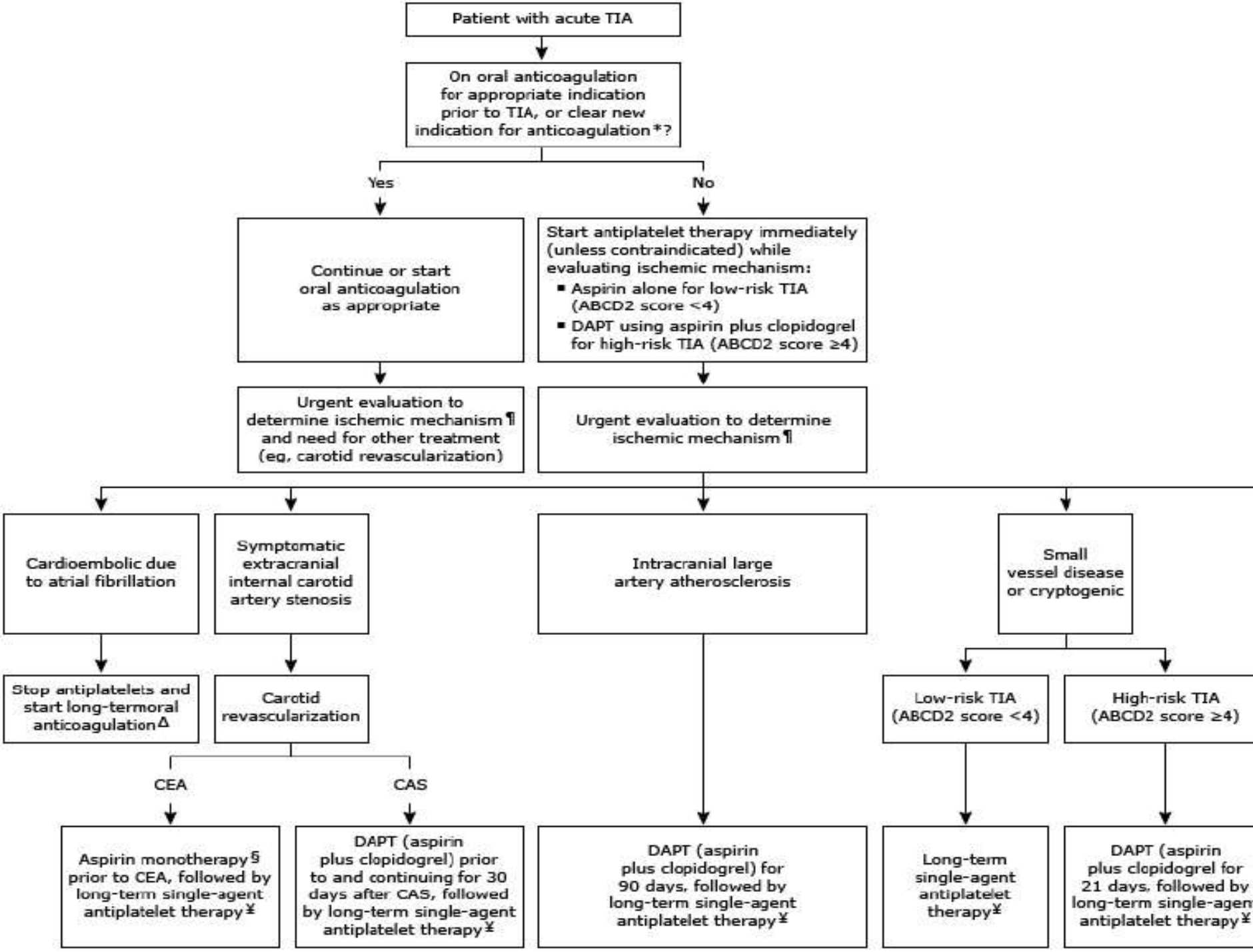
CARDIAC EVALUATION

- **ECG** should occur as soon as possible after TIA (**Class I, Level of EvidenceB**).
- Prolonged cardiac monitoring is useful in patients with an unclear origin afterinitial brain imaging and electrocardiography (**Class IIa, Levelof Evidence B**).
- Echocardiography (at least TTE) is reasonable in the evaluation of patients with suspected TIAs. (**Class IIa, Level of Evidence B**).
- TEE is useful in identifying patent foramen ovale (PFO), aortic arch atherosclerosis, and valvular disease and is reasonable when identification of these conditions will alter management (**Class IIa, Level of Evidence B**).

- Routine blood tests (CBC, chemistry panel, PT and aPPT, RFT, Fasting blood glucose and fasting lipid panel) are reasonable in the evaluation of patients with suspected TIAs (**Class IIa, Level of Evidence B**).

TREATMENT

TRANSIENT ISCHEMIC ATTACK


Patient with acute TIA
On oral anticoagulation for appropriate indication prior to TIA, or clear new indication for anticoagulation*?
Yes
No
Continue or start oral anticoagulation as appropriate
Start antiplatelet therapy immediately (unless contraindicated) while evaluating ischemic mechanism:
Aspirin alone for low-risk TIA (ABCD2 score <4)
DAPT using aspirin plus clopidogrel for high-risk TIA (ABCD2 score ≥4)
Urgent evaluation to determine ischemic mechanism¶ and need for other treatment (eg, carotid revascularization)
Urgent evaluation to determine ischemic mechanism¶
Cardioembolic due to atrial fibrillation
Symptomatic extracranial internal carotid artery stenosis
Intracranial large artery atherosclerosis
Small vessel disease or cryptogenic
Stop antiplatelets and start long-termoral anticoagulation Δ
Carotid revascularization
Low-risk TIA (ABCD2 score <4)
High-risk TIA (ABCD2 score ≥4)
CEA
CAS
Aspirin monotherapy§ prior to CEA, followed by long-term single-agent antiplatelet therapy¥
DAPT (aspirin plus clopidogrel) prior to and continuing for 30 days after CAS, followed by long-term single-agent antiplatelet therapy¥
DAPT (aspirin plus clopidogrel) for 90 days, followed by long-term single-agent antiplatelet therapy¥
Long-term single-agent antiplatelet therapy¥
DAPT (aspirin plus clopidogrel for 21 days, followed by long-term single-agen antiplatelet therapy¥

MANAGEMENT OF ACUTE ISCHEMIC STROKE

REPERFUSION THERAPIES

- The immediate goal of reperfusion therapy for acute ischemic stroke is to restore blood flow to the regions of brain that are ischemic but not yet infarcted.

- The long-term goal is to improve outcome by reducing stroke-related disability and mortality.

REPERFUSION THERAPIES

- **Alteplase** — tPA is the mainstay of treatment for acute ischemic stroke, provided that treatment is initiated within 4.5 hours of clearly defined symptom onset.
- Because the benefit of alteplase is time dependent, it is critical to treat patients as quickly as possible.
- Alteplase initiates local fibrinolysis by binding to fibrin in a thrombus (clot) and converting entrapped plasminogen to plasmin. In turn, plasmin breaks up the thrombus.

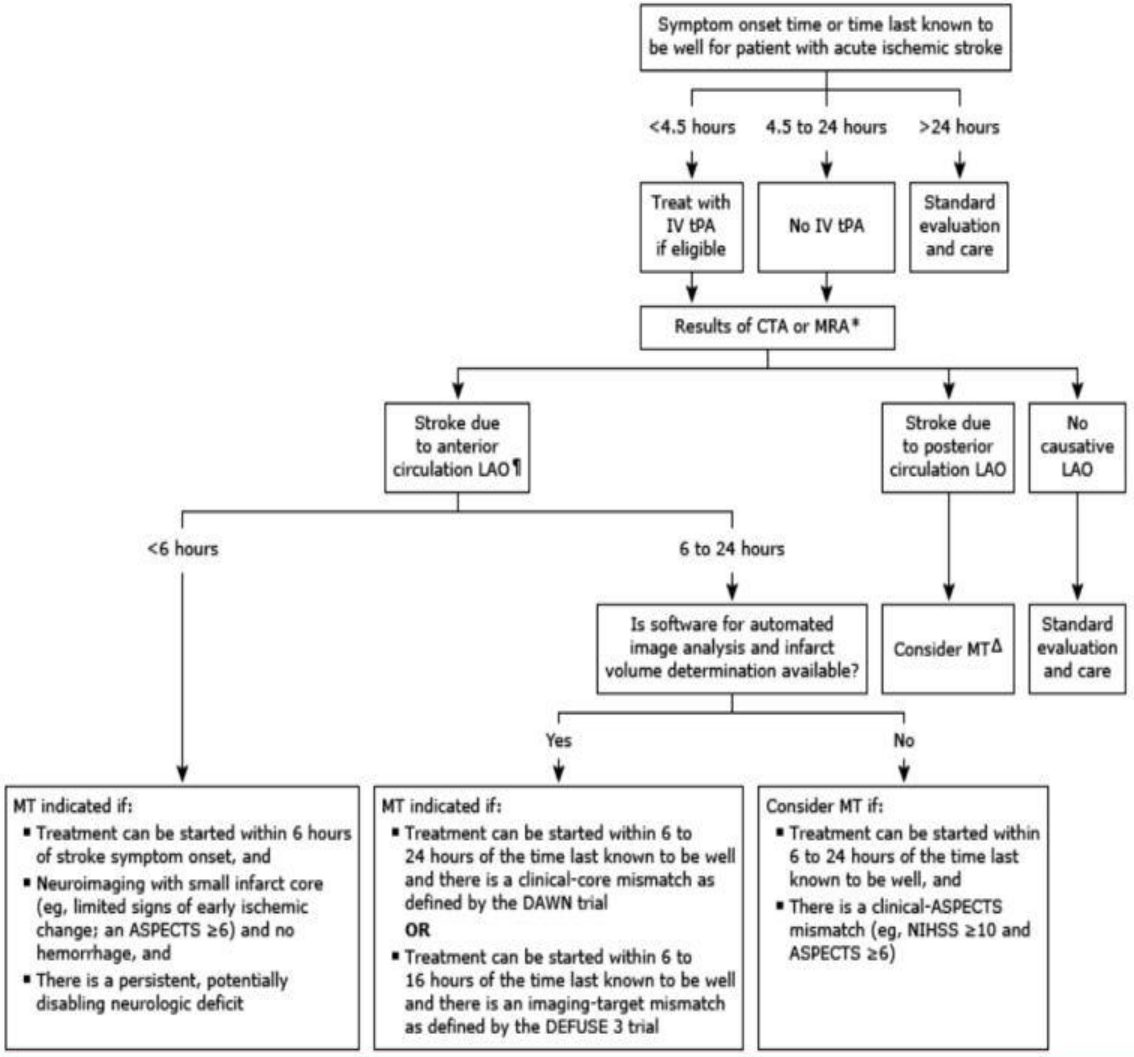
REPERFUSION THERAPIES

- **Tenecteplase** — Although not licensed in the US for thrombolysis in acute ischemic stroke treatment, there is moderate- to high-quality evidence that intravenous tenecteplase (0.25 mg/kg, maximum 25 mg, given in a single bolus) has similar efficacy and safety outcomes compared with alteplase, including rates of excellent functional outcome, symptomatic intracerebral hemorrhage, and mortality at 90 days.

REPERFUSION THERAPIES

- **Mechanical thrombectomy** — Mechanical thrombectomy is indicated for patients with acute ischemic stroke due to a large artery occlusion in the anterior circulation who can be treated within 24 hours of the time last known to be well (i.e., at neurologic baseline), regardless of whether they receive intravenous alteplase for the same ischemic stroke event.
- Patients with ischemic stroke from large vessel occlusion should receive intravenous alteplase without delay, if eligible, even if mechanical thrombectomy is being considered.
- Mechanical thrombectomy treatment should then be started as quickly as possible, and should not be delayed to assess the response to intravenous alteplase.
- Mechanical thrombectomy alone (without preceding intravenous thrombolysis) is an alternative strategy, but available data are inconclusive regarding the efficacy of this approach compared with the combination of intravenous thrombolysis and mechanical thrombectomy (ie, bridging therapy)

Indications for mechanical thrombectomy to treat patients with acute ischemic stroke



General criteria for MT require all of the following:

- Neuroimaging (eg, CT without contrast or diffusion-weighted MRI) is consistent with a small infarct core (ie, limited signs of early ischemic change) and excludes hemorrhage
- Angiography (eg, CTA or MRA) demonstrates a proximal LAO in the anterior circulation
- Thrombectomy is performed at a stroke center with appropriate expertise in MT
- The patient has a persistent, potentially disabling neurologic deficit
- Treatment can be started within 24 hours of the time last known to be well

DAWN trial eligibility require all of the following:

- Treatment (femoral puncture) can start within 6 to 24 hours of time last known to be well
- Failed or contraindicated for IV tPA
- A deficit on the NIHSS of ≥10 points
- No significant prestroke disability: Baseline mRS score ≤1
- Baseline infarct involving <1/3 of MCA territory
- Intracranial occlusion of the ICA or M1 segment of the MCA
- A clinical-core mismatch according to age:
  - Age ≥80 years: NIHSS ≥10 and infarct volume <21 mL
  - Age <80 years: NIHSS 10 to 19 and infarct volume <31 mL
  - Age <80 years: NIHSS ≥20 and infarct volume <51 mL

DEFUSE 3 trial eligibility require all of the following:

- Treatment (femoral puncture) can start within 6 to 16 hours of time last known to be well
- A deficit on the NIHSS of ≥6 points
- Only slight or no prestroke disability: Baseline mRS score ≤2
- Occlusion of the cervical or intracranial ICA (with or without tandem MCA lesions) or the M1 segment of the MCA
- Age 18 to 90 years
- A target mismatch profile on CT perfusion or MRI defined as:
  - An ischemic core volume <70 mL, and
  - A mismatch ratio (the volume of the perfusion lesion divided by the volume of the ischemic core) >1.8, and
  - A mismatch volume (volume of perfusion lesion minus the volume of the ischemic core) >15 mL

IV: intravenous; tPA: tissue plasminogen activator; CTA: computed tomography angiography; MRA: magnetic resonance angiography; LAO: large artery occlusion; MT: mechanical thrombectomy; ASPECTS: Alberta Stroke Program Early CT Score; NIHSS: National Institutes of Health Stroke Scale; CT: computed tomography; MRI: magnetic resonance imaging; mRS: modified Rankin Scale; MCA: middle cerebral artery; ICA: internal carotid artery.

* Usually assessed with MRA or CTA, less often with digital subtraction angiography.

¶ There is intracranial arterial occlusion of the distal ICA, middle cerebral (M1/M2), or anterior cerebral (A1/A2) artery by CTA, MRA, or digital subtraction angiography.

Δ MT may be a treatment option for patients with acute ischemic stroke caused by occlusion of the basilar artery, vertebral arteries, or posterior cerebral arteries at expert stroke centers, but benefit is uncertain.

INVESTIGATIONAL THERAPIES

- Intravenous desmoteplase or urokinase
- Intravenous glycoprotein IIb/IIIa receptor inhibitors
- Combinations of interventions (other than intravenous alteplase and mechanical thrombectomy) to restore perfusion

- Aspirin and other antithrombotic agents should not be given alone or in combination for the first 24 hours following treatment with intravenous tPA
- For patients without serious bleeding complications who are not on anticoagulation or antiplatelet therapy at baseline, antiplatelet therapy should be started as soon as possible (STUDIED WITHIN 12-24 HRS) while evaluating the ischemic stroke mechanism

Stroke. 2019;50:773-778

WHEN TO START SINGLE ANTI-PLATELET?

- Patients with moderate or higher stroke severity, defined by an National Institutes of Health Stroke Scale (NIHSS) score >3
- Monotherapy with Aspirin (162 to 325 mg/daily)
- Aspirin may be given rectally for patients with acute stroke who are NPO or those who have not had screening for dysphagia

Stroke. 2019;50:773-778

WHEN TO START DUAL ANTIPLATELET?

- Patients with minor ischemic stroke, defined by an NIHSS score ≤3
- Patients with stroke due to intracranial large artery atherosclerosis
- For patients on single antiplatelet therapy with aspirin or clopidogrel at the time of stroke onset(if NIHSS≤3)

Stroke. 2019;50:773-778

DURATION OF DUAL ANTIPLATELET THERAPY

- For **21 days** using aspirin (160 to 325 mg loading dose, followed by 50 to 100 mg daily) plus clopidogrel (300 mg loading dose, followed by 75 mg once daily)
- For **90 days** using aspirin (160 to 325 mg loading dose, followed by 50 to 100 mg daily) plus clopidogrel (300 mg loading dose, followed by 75 mg once daily) for patients with stroke due **to intracranial large artery atherosclerosis**

Stroke. 2019;50:773-778

TRIPLE ANTIPLATELET THERAPY

- Triple antiplatelet therapy for acute stroke may be harmful, even with short-term use
- **Not** reduce the incidence or severity of recurrent stroke or TIA
- Increase risk of major bleeding

Stroke. 2019;50:773-778

ROLE OF TICAGRELOR IN OUTCOME AFTER STROKE

SOCRATES	13 199 patients with nonsevere ischemic stroke or high-risk TIA within 24 h	Ticagrelor (180 mg load then 90 mg BID) vs aspirin (300 mg load then 100 mg/d)	90 d	HR=0.89 (*P*=0.07) for stroke/MI/death; HR=0.87 (*P*=0.046) for ischemic stroke; HR=0.83 (*P*=0.45) for major bleeding

- Ticagrelor monotherapy was not significantly better than aspirin monotherapy (both started within 24 hours of symptom onset) for the 90-day composite endpoint of stroke, myocardial infarction, or death

- Prespecified exploratory analysis(Amarenco et al), ticagrelor was superior to aspirin in the subgroup of patients who had stroke of possible atherosclerotic origin (defined as ipsilateral atherosclerotic stenosis of an extracranial or intracranial artery (including <50 percent stenosis), or mobile thrombus or thick plaque (≥4 mm) in the aortic arch

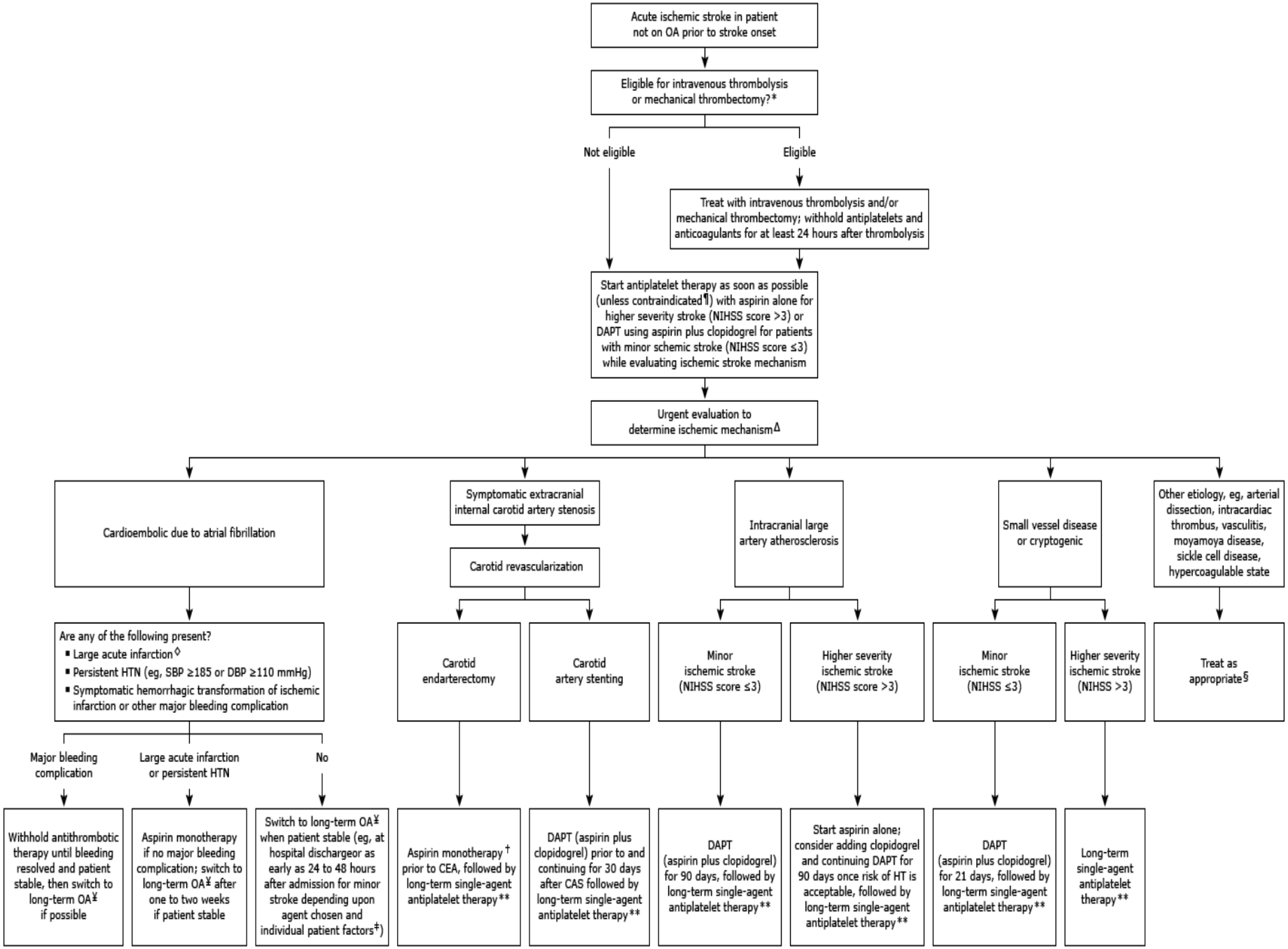
Stroke. 2019;50:773-778

Selection of anti-platelets for secondary prevention of stroke

- For non-cardioembolic ischemic stroke or transient ischemic attack (TIA) of atherothrombotic, lacunar(small vessel occlusive type), or cryptogenic type, treatment with an antiplatelet agent using either aspirin, clopidogrel, or aspirin-extended-release dipyridamole is recommended

Stroke.
2014;45:3754-3832

General approach to antithrombotic therapy for acute ischemic stroke in patient not on anticoagulation prior to stroke onset



ROLE OF ANTICOAGULATION IN STROKE

IN PATIENTS WITH TIA

- For patients with a clear indication for anti-coagulation,

example-

- Atrial fibrillation
- venous thromboembolism
- mechanical heart valve

- Anticoagulation rather than antiplatelet therapy is recommended

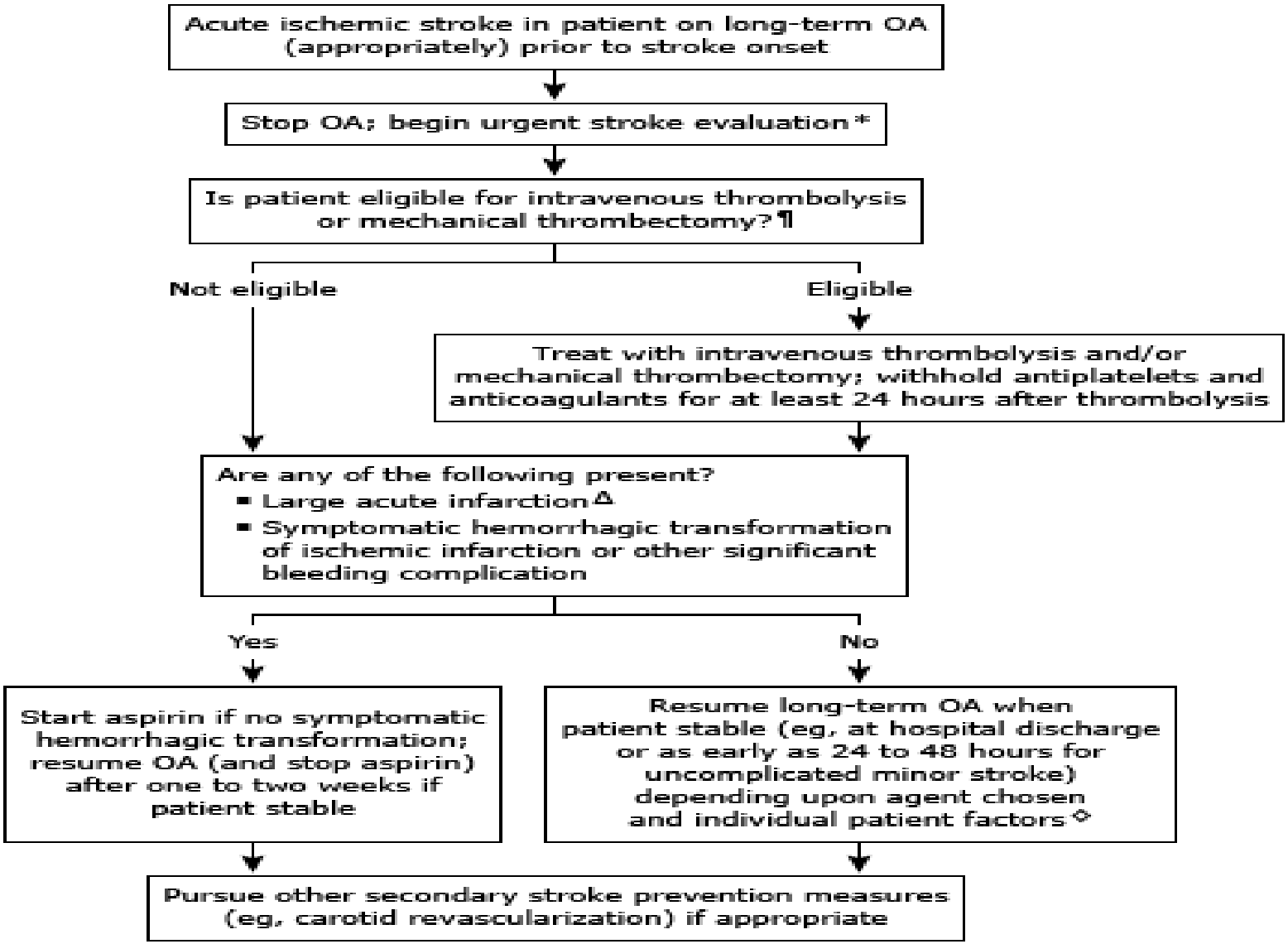
J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

- In patients who are sub-therapeutically or not anti-coagulated at presentation, **bridging anticoagulation** with heparin, low molecular weight heparin, or a direct oral anticoagulant (DOAC) should be considered
- In patients who are therapeutically anticoagulated at presentation, management should be individualized based on the underlying mechanism of the TIA
- In some instances, such as when the **TIA is more likely due to atherosclerosis** than to cardioembolism, it may be reasonable to add single antiplatelet therapy

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

PATIENTS WITH ACUTE ISCHEMIC STROKE

General approach to antithrombotic therapy for acute ischemic stroke in patient on long-term anticoagulation (appropriately) prior to stroke onset



CARDIOEMBOLIC SOURCE

Atrial fibrillation

- Parenteral anticoagulation is generally **not** recommended for treating ischemic stroke in the acute period
- Oral anticoagulation with warfarin or a DOAC is recommended for secondary stroke prevention in patients with atrial fibrillation and other high-risk sources of cardiogenic embolism

Intra-cardiac thrombus

- Early parenteral anticoagulation rather than aspirin only is recommended for select patients with acute cardioembolic ischemic stroke who are at high risk for short-term recurrent stroke due to intracardiac thrombus in the left ventricle or thrombus associated with mechanical or native heart valves
- Transitioned to oral anticoagulation soon after initiation of parenteral anticoagulation is indicated

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

- Oral anticoagulation can be started **immediately** for patients with **TIA**
- Medically stable patients with **a small or moderate-sized infarct** and no bleeding complications

➢ VKA-within24 hrs/by the time of discharge

➢ NOAC –after 48 hrs

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

- **Withholding oral anticoagulation for one to two weeks is generally recommended**

➢ patients with large infarctions

➢ symptomatic hemorrhagic transformation, or poorly controlled hypertension(eg, systolic blood pressure ≥185 or diastolic blood pressure ≥110 mmHg)

➢ Or bleeding complications

- Aspirin to be started if there are no significant bleeding complications
- Anticoagulation can be resumed (and aspirin stopped) after one to two weeks if the patient is stable

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

- Oral anticoagulation is generally continued at least **three months** for left ventricular thrombus
- **Concurrent antiplatelet therapy** may be indicated for left ventricular thrombus after acute myocardial infarction
- **Chronic anticoagulation** is indicated for patients with mechanical heart valves

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

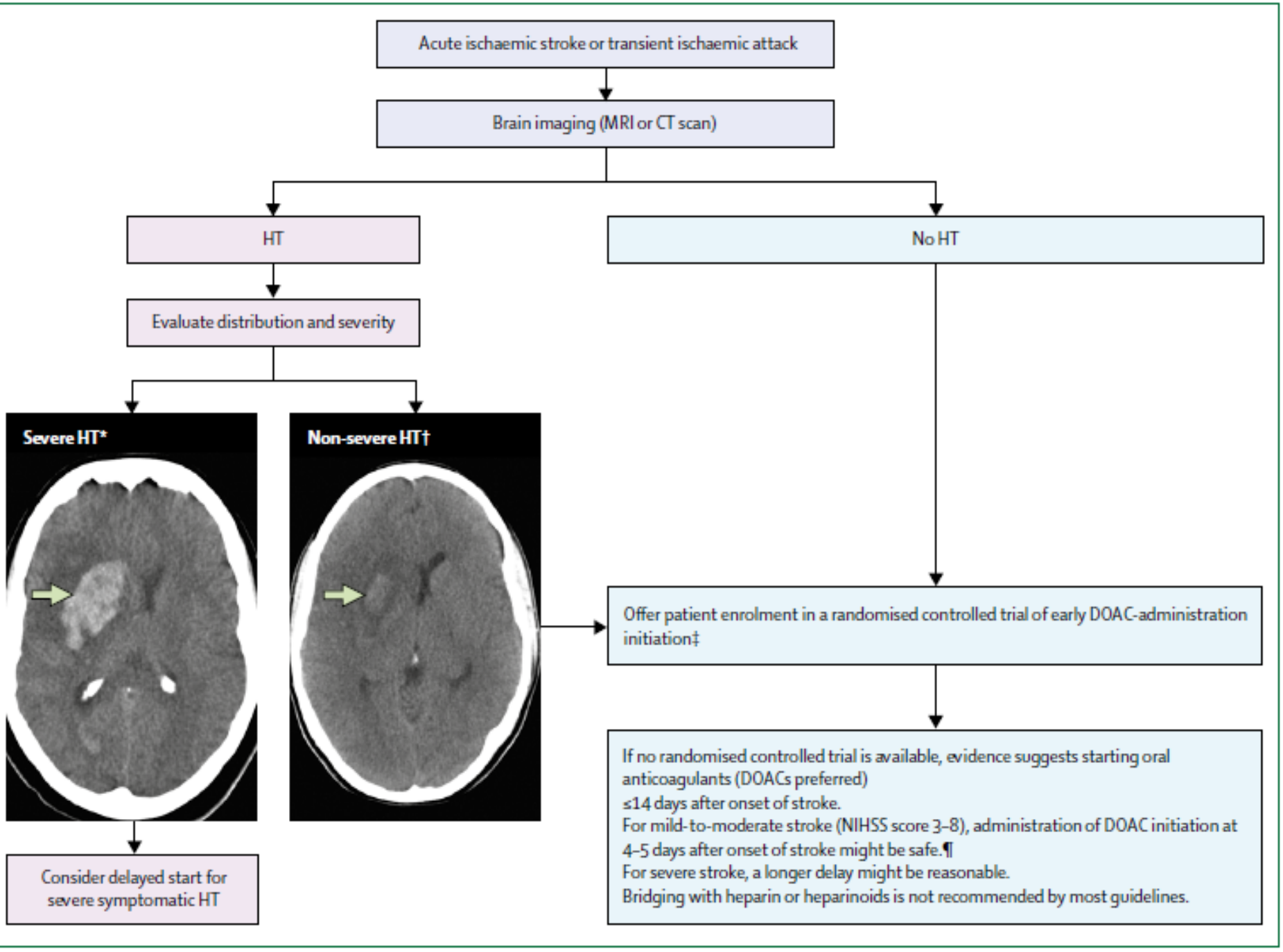


Figure 2: **Timing for initiation of direct oral anticoagulant administration**

Anti-coagulant of choice

Anticoagulation with NOACs led to similar or lower rates both of ischemic stroke and major bleeding compared to adjusted dose warfarin(INR of 2.0 to 3.0) in patients with non-valvular AF

The following are situations in which it is reasonable or even necessary to prefer warfarin instead of the NOAC agents:

- Patients already on warfarin who are comfortable with periodic international normalized ratio (INR) measurement
- Patients with mechanical heart valves of any type or those with severe mitral stenosis of any cause

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

- Patients who are not likely to comply with the twice daily dosing of dabigatran or apixaban
- Patients for whom the NOAC agents will lead to an unacceptable increase in cost
- Patients with chronic severe kidney disease whose estimated glomerular filtration rate is less than 30 mL/min(apixaban is approved for use in the United States for patients with end stage renal disease)
- Patients for whom the NOAC agents are contraindicated, including those on enzyme-inducing antiepileptic drugs (eg, phenytoin) and patients with human immunodeficiency virus infection (HIV) on protease inhibitor-based antiretroviral therapy

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

Important studies supporting the evidence

- A 2014 **Cochrane review** compared the factor Xa inhibitors (apixaban, betrixaban, darexaban, edoxaban, idraparinux, and rivaroxaban) to warfarin in patients with AF and found a lower rate of stroke and systemic embolic events with the former (OR 0.81, 95% CI 0.72-0.91; absolute rates 2.5 versus 3.2 patients, respectively), as well a lower rate of death and ICH
- Meta-analyses with pooled the results from the **RE-LY** (dabigatran) , **ARISTOTLE** (apixaban) , and **ROCKET AF** (rivaroxaban) trials and reached to the conclusions of superiority of NOAC over warfarin

- The NOAC agents (compared to warfarin) are associated with the following:

- A **significant reduction of stroke/systemic embolism** (odds ratio [OR] 0.85, 95% confidence interval [CI] 0.74-0.99; absolute risk reduction, 0.7 percent) and major bleeding (OR 0.86, 95% CI 0.75-0.99; absolute risk reduction 0.8 percent)
- A **significant and marked relative reduction in hemorrhagic stroke** (relative risk [RR] 0.48, 95% CI 0.36-0.62) and a significant reduction in all-cause mortality (RR 0.88, 95% CI 0.82-0.96)
- In these meta-analyses, there was a trend **toward reduced major bleeding with the NOAC agents** (relative risk 0.86, 95% CI 0.72-1.02 and 0.80, 95% CI 0.63-1.01)

Anticoagulant overlap

- For overlapping a heparin product with a VKA co-administer the two anticoagulants until the effect of the VKA is established
- For warfarin, this generally entails four to five days of overlap and at least 24 hours of a therapeutic international normalized ratio (INR)
- For direct-acting oral anticoagulants (DOACs) the maximal anticoagulant effect of the oral anticoagulant is achieved within two to three hours of the first dose. Thus, intravenous unfractionated heparin can be stopped when the first dose of the DOAC is taken

J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management

NEUROPROTECTIVE AGENTS

3.12. Neuroprotective Agents

3.12. Neuroprotective Agents	COR	LOE	New, Revised, or Unchanged
1. At present, pharmacological or nonpharmacological treatments with putative neuroprotective actions are not recommended.	III: No Benefit	A	Recommendation reworded for clarity from 2013 AIS Guidelines. LOE unchanged. COR amended to conform with ACC/AHA 2015 Recommendation Classification System. See Table XCV in online Data Supplement 1 for original wording.

3.1. Airway, Breathing, and Oxygenation (Continued)	COR	LOE	New, Revised, or Unchanged
4. Hyperbaric oxygen (HBO) is not recommended for patients with AIS except when caused by air embolization.	III: No Benefit	B-NR	Recommendation revised from 2013 AIS Guidelines.
2. In patients with AIS, the benefit of treatment with induced hypothermia is uncertain.	IIb	B-R	Recommendation revised from 2013 AIS Guidelines.

6.8.4. Timing	COR	LOE	New, Revised, or Unchanged
1. Among patients already taking statins at the time of onset of ischemic stroke, continuation of statin therapy during the acute period is reasonable.	IIa	B-R	Recommendation and COR unchanged from 2013 AIS Guidelines. LOE amended to conform with ACC/AHA 2015 Recommendation Classification System.
2. For patients with AIS who qualify for statin treatment, in-hospital initiation of statin therapy is reasonable.	IIa	C-LD	New recommendation.

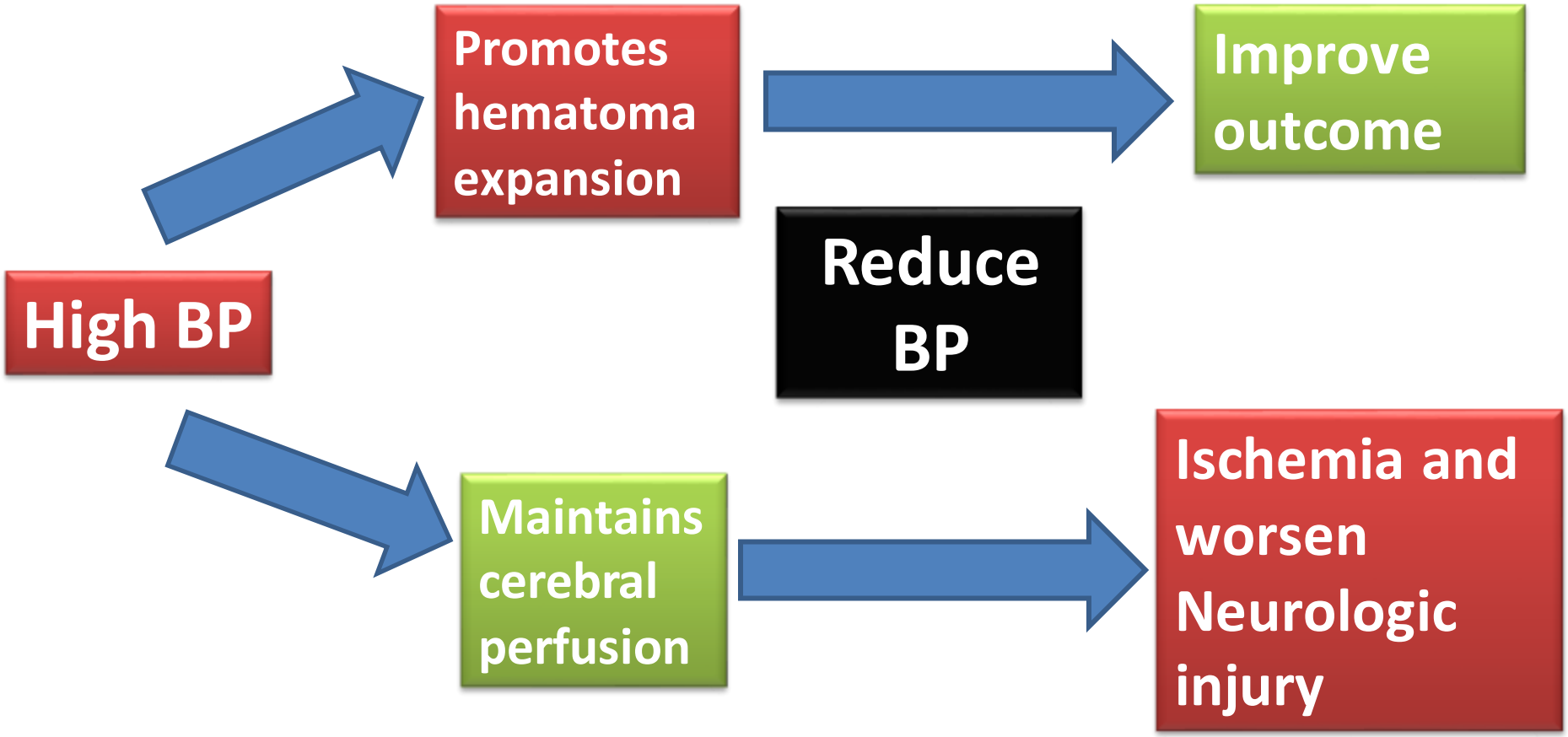
American Stroke Association.

INTRACEREBRAL HEMORRHAGE

Clinical indicators for ICH

- Vomiting
- Systolic BP (SBP) >220 mm Hg
- Severe headache
- Coma or decreased level of consciousness,
- Symptom progression over minutes or hours

BP management: A double edged sword



BP management in ICH

CLASS I, LOE A : **For ICH patients presenting with SBP between 150 and 220 mm Hg and without contraindication to acute BP treatment, acute lowering of SBP to 140 mm Hg is safe.**

CLASS IIB, LOE C: **For ICH patients presenting with SBP >220 mm Hg, it may be reasonable to consider aggressive reduction of BP with a continuous intravenous infusion and frequent BP monitoring**

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Management of Intracranial pressure

BASIC measures:

- Elevation of the head of the bed to 30 degrees
- Mild sedation, as needed for comfort
- Use of normal saline initially for maintenance and replacement fluids; hypotonic fluids are contraindicated
- Glucocorticoids : **NOT** to be used in ICH
- CPP target : 50-70mm Hg

Management of Intracranial pressure

Osmotic therapy:

- Mannitol : Initial bolus of **0.5 to 1 g/kg** → Maximum of **250 mg/kg** 4th hourly based on need
- Monitoring of serum osmolality should be done
- The goal of therapy is to achieve plasma hyperosmolality (300 to 310 mosmol/kg) to force water to exit the brain
- Mannitol dose should not exceed **250 mg/kg** every 4 hours; higher doses can cause reversible acute renal failure.

Management of Intracranial pressure

- Hypertonic saline: 3% NaCl may be titrated to a sodium goal of approximately 145 to 155 mEq/L.
- Salvage therapy: Propofol is given intravenously to ventilated patients with a loading dose of 1 to 3 mg/kg and continued as an infusion with titration to achieve the desired sedation level, typically at 5 to 50 mcg/kg per minute
- Hyperventilation: A partial pressure of carbon dioxide ($PaCO_2$) goal of 30 to 35 mmHg is suggested

Glucose management

CLASS I, LEVEL B: Glucose should be monitored. Both hyperglycemia and hypoglycemia should be avoided

- High blood glucose on admission predicts an increased risk of mortality and poor outcome in patients with ICH, independent of the presence of DM

- No fixed glycemic targets

- RCT on maintaining glucose at 80-110, showed higher mortality risk and hypoglycemic events

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Seizure management

- The frequency of clinical seizures early (within 1 week) after ICH around 16%
- Majority occurring at or near onset
- Patients with a change in mental status who are found to have electrographic seizures on EEG should be treated with AEDs

CLASS III, LOE C: Prophylactic antiseizure medication is not recommended

CLASS IIA, LOE- C Continuous EEG monitoring is indicated in ICH patients with depressed mental status that is out of proportion to the degree of brain injury

Management of medical complications

The most common complications of ICH include:

- Pneumonia (5.6%)
- Aspiration (2.6%)
- Respiratory failure/distress (2%)
- PE (1.3%), and
- Sepsis (1.7%).

CLASS I, LOE B: A formal screening procedure for dysphagia should be performed in all patients before the initiation of oral intake to reduce the risk of pneumonia

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Management of medical complications

CLASS I, LOE B: Systematic screening for myocardial ischemia or infarction with electrocardiogram and cardiac enzyme testing after ICH is reasonable

- In a meta-analysis of 65 996 stroke patients with a mean follow-up of ≈3.5 years, the annual risk of MI was 2.2 %.
- An elevated troponin level >0.4 ng/mL was found in 15% within 24 hours of admission and was associated with increased in-hospital mortality
- Heart failure can result from myocardial ischemia, cardiomyopathy (stress induced)
- Neurogenic pulmonary edema: Ventilator support

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

The end to Statin controversy

Cerebrovascular disease

RESEARCH PAPER

Statins and the risk of intracerebral haemorrhage in patients with stroke: systematic review and meta-analysis

Oliver Jonathan Ziff,[1,2] Gargi Banerjee,[1] Gareth Ambler,[3] David J Werring[1,2]

ABSTRACT

Objective Whether statins increase the risk of intracerebral haemorrhage (ICH) in patients with a previous stroke remains uncertain. This study addresses the evidence of statin therapy on ICH and other clinical outcomes in patients with previous ischaemic stroke (IS) or ICH.

Methods A systematic literature review and meta-analysis was performed in conformity with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines to assess observational and randomised studies comparing statin therapy with control (placebo or no treatment) in patients with a previous ICH or IS. The risk ratios (RR) for the primary outcome (ICH) and secondary outcomes (IS, any stroke, mortality and function) were pooled using random effects meta-analysis according to stroke subtype.

Results Forty-three studies with a combined total of 317 291 patient-years of follow-up were included. In patients with previous ICH, statins had no significant impact on the pooled RR for recurrent ICH (1.04, 95% CI 0.86 to 1.25; n=23 695); however, statins were associated with significant reductions in mortality (RR 0.49, 95% CI 0.36 to 0.67; n=89 976) and poor functional outcome (RR 0.71, 95% CI 0.67 to 0.75; n=9113). In patients with previous IS, statins were associated with a non-significant increase in ICH (RR 1.36, 95% CI 0.96 to 1.91; n=103 525), but significantly lower risks of recurrent IS (RR 0.74, 95% CI 0.66 to 0.83; n=53 162), any stroke (RR 0.82, 95% CI 0.67 to 0.99; n=55 260), mortality (RR 0.68, 95% CI 0.50 to 0.92; n=74 648) and poor functional outcome (RR 0.83, 95% CI 0.76 to 0.91; n=34 700).

Conclusions Irrespective of stroke subtype, there were non-significant trends towards future ICH with statins. However, this risk was overshadowed by substantial and significant improvements in mortality and functional outcome among statin users.

Ziff OJ et at. Statins and the risk of intracerebral haemorrhage in patients with stroke: systematic review and meta-analysis, JNNP, 2018

The end to Statin controversy

A new meta-analysis has concluded that the benefit of statin therapy in the prevention of Ischemic stroke "greatly exceeds" the risk for ICH. The meta-analysis was released on AAN.com as part of the 2020 American Academy of Neurology Science Highlights.

Does the Pre- or Post-stroke Statin Therapy Increase the Risk of Intracerebral Hemorrhage in Post-stroke Patients? A Meta-analysis of Randomized and Nonrandomized Studies

Ammad Ishfaq[1], Muhammad Ishfaq[2], Fnu Abhi Pandhi[3], Aman Deep[4], Ana Hosseinzadeh Maleki[5], Reza Bavarsad Shahripour[5]
[1]Neurology, Zeenat Qureshi Stroke Institute, [2]Department of Neurology, UTHSC, [3]UTHSC Dept of Neurology, [4]University of Tennessee Health Science Center Neurology, [5]University of Tennessee

Objective:
The role of statin in primary and secondary prevention and the risk of intracerebral hemorrhage.

Background:

The association of statin and intracerebral hemorrhage risk is controversial. We conducted this meta-analysis to assess the safety of statin by estimating the risk versus benefit in terms of cerebral hemorrhagic and ischemic events.

Design/Methods:

A total of 19 clinical studies met the inclusion criteria and were included in the meta-analysis. We calculated pooled risk ratios (RR) and 95% confidence intervals (CI) using random-effects models. The primary endpoint was intracerebral hemorrhage. Secondary endpoints were ischemic stroke and transient ischemic attack.

Results:

A total of 35,842 were included in this metanalysis. The risk of combined primary and secondary intracerebral hemorrhage was not significantly associated with statin (RR 1.03; 95% CI 0.85-1.08). Sensitivity analysis showed a trend toward a higher risk of secondary intracerebral hemorrhage among those who were assigned to statin treatment (OR 1.87, 95% CI .91-3.86). The risk of cerebral ischemia (stroke and transient ischemic attack) was significantly lower in subjects assigned to statin treatment (*RR* 0.79; 95% CI 0.61-0.87).

Conclusions:

The benefits of lipid-lowering therapy in the prevention of ischemic stroke greatly exceed the risk of ICH.

Surgical treatment in ICH

Resumption of Anti-platelets and Anti-coagulation

- Antiplatelets (when indicated) typically should not be restarted until at least **one to two weeks** after the onset of ICH
- Aspirin can be used safely within days, as soon as **48 hours**, after ICH in stable patients who require it
- In the **RESTART** trial, the benefit of aspirin was found to be more prominent among patients with non-lobar ICH than among those with lobar ICH
- Delaying oral anticoagulants for at least **four weeks** after onset of the ICH

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Prevention of Recurrent ICH

Risk factors for recurrence:

(1) Lobar location of the initial ICH;
(2) Uncontrolled hypertension
(3) Older age
(4) Presence and number of microbleeds on gradient echo MRI;
(5) Ongoing anticoagulation; and
(6) Presence of apolipoprotein E ε2 or ε4 alleles
(7) Male
(8) Ischemic stroke in the past
(9) Black race

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Prevention of Recurrent ICH: Recommendations

- Avoidance of oral anticoagulation for at least 4 weeks, in patients without mechanical heart valves, might decrease the risk of ICH recurrence

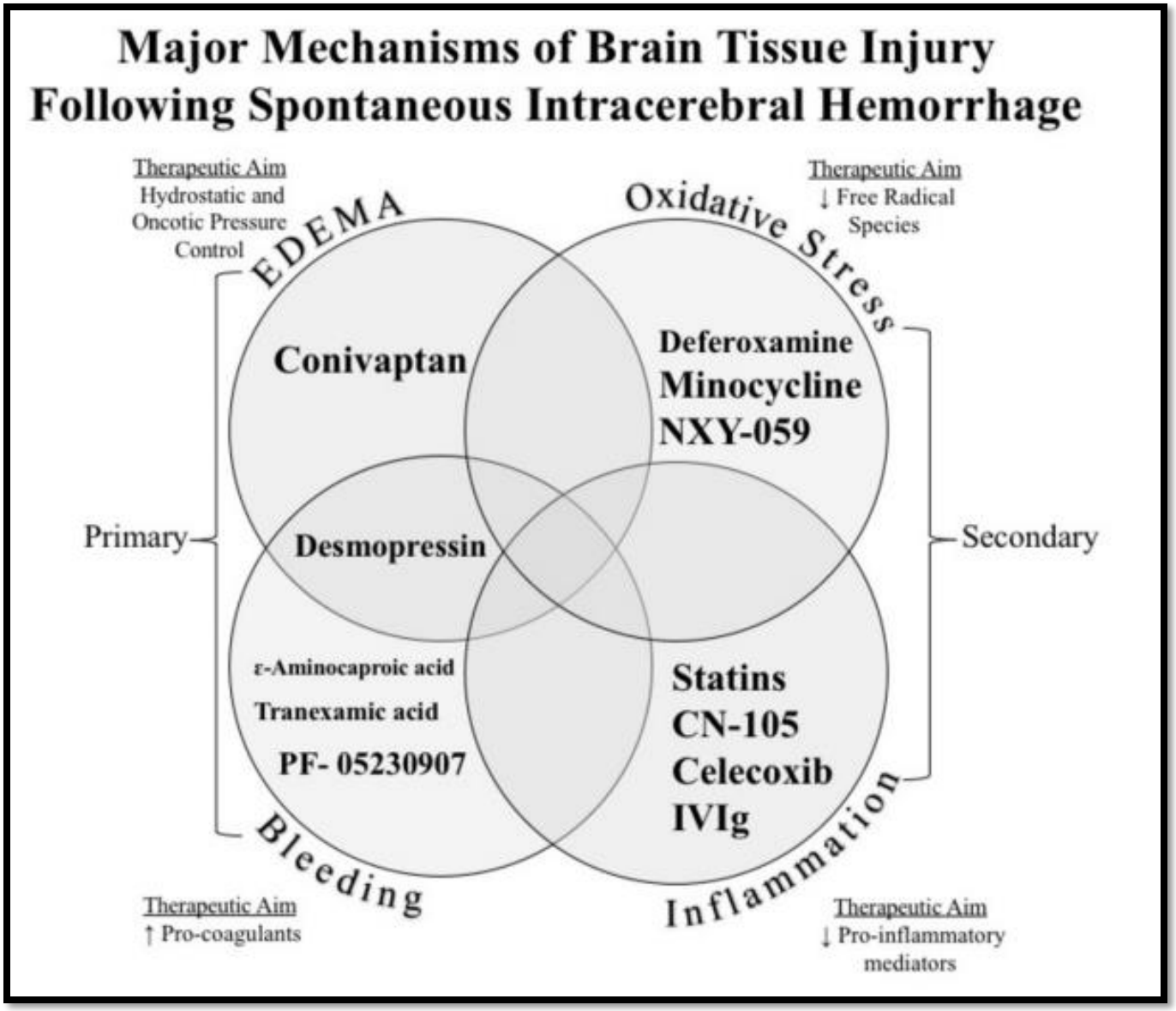
- Lifestyle modifications, including avoidance of alcoholuse greater than 2 drinks per day, tobacco use, and illicit drug use, as well as treatment of obstructive sleep apnea, are probably beneficial

Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, AHA, July 2015

Prevention of Recurrent ICH: Recommendations

- Long-term anticoagulation with warfarin as a treatment for nonvalvular atrial fibrillation is not recommended after warfarin-associated spontaneous lobar ICH because of the relatively high risk of recurrence
- The usefulness of dabigatran, rivaroxaban, or apixaban in patients with atrial fibrillation and past ICH to decrease the risk of recurrence is uncertain
- There are insufficient data to recommend restrictions on the use of statins in ICH patients .
- Antiplatelet agents do not appear to dramatically increase the risk of hematoma expansionand therefore appear to be generally safe for use after ICH, including ICH caused by CAA.

Recent advances



Babadjouni M, Neuroprotective strategies following intraparenchymal hemorrhage, BMJ 2017

Mortality and Functional outcome

Predictors of mortality and poor functional outcome after ICH include:

- Increasing age
- Decreasing Glasgow Coma Scale (GCS) score
- Increasing ICH volume
- Presence of intraventricular hemorrhage
- Deep or infratentorial ICH location
- Preceding oral anticoagulation therapy, and possibly antiplatelet therapy

Mortality and Functional outcome

Additional factors predicting mortality:

- Elevated **admission blood glucose** however, it is unclear if elevated glucose directly contributes to poor outcome, or if it alternatively is present secondarily as part of the stress response to severe ICH
- **Lower** serum levels of low density lipoprotein cholesterol (LDL-C) ; **low** total cholesterol and low LDL-C have also been linked to a risk of ICH
- Elevated C-reactive protein levels
- Extensive white matter lesions on CT or magnetic resonance imaging (MRI)

Venous thromboembolism

- Incidence of pulmonary embolism in first few months – 1-3%
- Early death after stroke due to pulmonary embolism – 13-25%
- Fatal embolism are seen most commonly encountered in weeks 2 to 4
- Peak incidence of DVT between 2-7 days

Venous thromboembolism

- Risk factor for DVT

- Severity of impaired mobility
- Dehydration
- Advanced age
- Malignancy
- Prior history of DVT
- Clotting disorders

4.8. Deep Vein Thrombosis Prophylaxis	COR	LOE	New, Revised, or Unchanged
1. In immobile stroke patients without contraindications, intermittent pneumatic compression (IPC) in addition to routine care (aspirin and hydration) is recommended over routine care to reduce the risk of deep vein thrombosis (DVT).	I	B-R	Recommendation revised from 2016 Rehab Guidelines.
confirmed these results.			
2. The benefit of prophylactic-dose subcutaneous heparin (unfractionated heparin [UFH] or LMWH) in immobile patients with AIS is not well established.	IIb	A	New recommendation.
The most recent and comprehensive meta-analysis of pharmacological interventions for venous thromboembolism			See Table LVI in online Data Supplement 1.

4.8. Deep Vein Thrombosis Prophylaxis (Continued)	COR	LOE	New, Revised, or Unchanged
3. When prophylactic anticoagulation is used, the benefit of prophylactic-dose LMWH over prophylactic-dose UFH is uncertain.	IIb	B-R	New recommendation.
4. In ischemic stroke, elastic compression stockings should not be used.	III: Harm	B-R	Recommendation wording modified from 2016 Rehab Guidelines to match Class III stratifications. COR and LOE amended to conform with ACC/AHA 2015 Recommendation Classification System.

Physical therapy is started at the bedside - flex and extend the knees and ankles throughout the day.

Infection

- Fever after stroke – Prompt evaluation for common sources including pneumonia, UTI
- Most common cause of fever in first 48 hours after acute stroke is pneumonia, which is attributed to aspiration in 60% of cases.
- Aspiration – decreased alertness, dysphagia and other neurological deficits like apraxia, agnosia, depression.

Infection - Aspiration

- In recumbent position – atelectasis and poor mobilisation of secretions.
- Coughing and deep breathing may not be performed.
- Decreased chest movement, decreased respiratory drive and function of interstitial muscles often abnormal on hemiplegic side. Kaldor and berlin noted pneumonia on hemiplegic side.

Kaldor A, Berlin I: Pneumonia, stroke, and laterality. Lancet 1981;1:843.

Review article

Use of tube feeding to prevent aspiration pneumonia

Thomas E Finucane, Julie P W Bynum

Conclusion

Although the use of feeding tubes is widespread, no data show that they reduce the risk of aspiration pneumonia in neurogenic dysphagia. There are data to the contrary, and the burdens of feeding tube placement can be severe. For almost all conscious patients we suggest a dedicated attempt at feeding by hand. We would generally reserve the offer of enteral tube feeding, as prophylaxis against aspiration and pneumonia, to those who have developed recurrent pneumonia despite our best efforts, those whose coughing during meals is extremely uncomfortable, and those who are acutely ill with impaired consciousness.

The Preventive Antibiotics in Stroke Study (PASS): a pragmatic randomised open-label masked endpoint clinical trial

Willeke F Westendorp*, Jan-Dirk Vermeij*, Elles Zock, Imke J Hooijenga, Nyika D Kruyt, Hans J L W Bosboom, Vincent I H Kwa, Martijn Weisfelt, Michel J M Remmers, Robert ten Houten, A H C M (Tobien) Schreuder, Sarah E Vermeer, Ewout J van Dijk, Diederik W J Dippel, Marcel G W Dijkgraaf, Lodewijk Spanjaard, Marinus Vermeulen, Tom van der Poll, Jan M Prins, Frederique H Vermeij, Yvo B W E M Roos, Ruud P Kleyweg, Henk Kerkhoff, Matthijs C Brouwer, Aeilko H Zwinderman, Diederik van de Beek†, Paul J Nederkoorn†, for the PASS investigators

Summary

Background In adults with acute stroke, infections occur commonly and are associated with an unfavourable functional outcome. In the Preventive Antibiotics in Stroke Study (PASS) we aimed to establish whether or not preventive antimicrobial therapy with a third-generation cephalosporin, ceftriaxone, improves functional outcome in patients with acute stroke.

Methods In this multicentre, randomised, open-label trial with masked endpoint assessment, patients with acute stroke were randomly assigned to intravenous ceftriaxone at a dose of 2 g, given every 24 h intravenously for 4 days, in addition to stroke unit care, or standard stroke unit care without preventive antimicrobial therapy; assignments were made within 24 h after symptom onset. The primary endpoint was functional outcome at 3 months, defined according to the modified Rankin Scale and analysed by intention to treat. The primary analysis was by ordinal regression of the primary outcome. Secondary outcomes included death, infection rates, antimicrobial use, and length of hospital stay. Participants and caregivers were aware of treatment allocation but assessors of outcome were masked to group assignment. This trial is registered with controlled-trials.com, number ISRCTN66140176.

Findings Between July 6, 2010, and March 23, 2014, a total of 2550 patients from 30 sites in the Netherlands, including academic and non-academic medical centres, were randomly assigned to the two treatment groups: 1275 patients to ceftriaxone and 1275 patients to standard treatment (control group). 12 patients (seven in the ceftriaxone group and five in the control group) withdrew consent immediately after randomisation, leaving 2538 patients available for the intention-to-treat-analysis (1268 in the ceftriaxone group and 1270 in the control group). 2514 (99%) of 2538 patients (1257 in each group) completed 3-month follow-up. Preventive ceftriaxone did not affect the distribution of functional outcome scores on the modified Rankin Scale at 3 months (adjusted common odds ratio 0·95 [95% CI 0·82–1·09], p=0·46). Preventive ceftriaxone did not result in an increased occurrence of adverse events. Overgrowth infection with *Clostridium difficile* occurred in two patients (<1%) in the ceftriaxone group and none in the control group.

Interpretation Preventive ceftriaxone does not improve functional outcome at 3 months in adults with acute stroke. The results of our trial do not support the use of preventive antibiotics in adults with acute stroke.

Prophylactic antibiotics after acute stroke for reducing pneumonia in patients with dysphagia (STROKE-INF): a prospective, cluster-randomised, open-label, masked endpoint, controlled clinical trial

Lalit Kalra, Saddif Irshad, John Hodsoll, Matthew Simpson, Martin Gulliford, David Smithard, Anita Patel, Irene Rebollo-Mesa, on behalf of the STROKE-INF Investigators*

Summary

Background Post-stroke pneumonia is associated with increased mortality and poor functional outcomes. This study assessed the effectiveness of antibiotic prophylaxis for reducing pneumonia in patients with dysphagia after acute stroke.

Methods We did a prospective, multicentre, cluster-randomised, open-label controlled trial with masked endpoint assessment of patients older than 18 years with dysphagia after new stroke recruited from 48 stroke units in the UK, accredited and included in the UK National Stroke Audit. We excluded patients with contraindications to antibiotics, pre-existing dysphagia, or known infections, or who were not expected to survive beyond 14 days. We randomly assigned the units (1:1) by computer to give either prophylactic antibiotics for 7 days plus standard stroke unit care or standard stroke unit care only to patients clu[stered] the units within 48 h of stroke onset. We did the randomisation with minimisation to stratify for numb[ers] and access to specialist care. Patient and staff who did the assessments and analyses were m[asked] ation. The primary outcome was post-stroke pneumonia in the first 14 days, assessed [by] archical algorithm and by physician diagnosis in the intention-to-treat popu[lation] intention to treat. This trial is closed to new participants and is registered with [ISRCTN] 18456.

Findi[ngs] [Between] ay 17, 2014, we randomly assigned 48 stroke units (and 1224 patients clustered [in] ment groups: 24 to antibiotics and 24 to standard care alone (control). 11 units and [after] er randomisation before 14 days, leaving 1217 patients in 37 units for the intention-to-treat [analysis] in the antibiotics group, 602 in control). Prophylactic antibiotics did not affect the incidence of [confirmed] post-stroke pneumonia (71 [13%] of 564 patients in antibiotics group *vs* 52 [10%] of 524 in control [group]; marginal adjusted odds ratio [OR] 1·21 [95% CI 0·71–2·08], p=0·489, intraclass correlation coefficient [ICC] [0·07] [95% CI 0·02–0·17]. Algorithm-defined post-stroke pneumonia could not be established in 129 (10%) patients because of missing data. Additionally, we noted no differences in physician-diagnosed post-stroke pneumonia between groups (101 [16%] of 615 patients *vs* 91 [15%] of 602, adjusted OR 1·01 [95% CI 0·61–1·68], p=0·957, ICC 0·08 [95% CI 0·03–0·21]). The most common adverse events were infections unrelated to post-stroke pneumonia (mainly urinary tract infections), which were less frequent in the antibiotics group (22 [4%] of 615 *vs* 45 [7%] of 602; OR 0·55 [0·32–0·92], p=0·02). Diarrhoea positive for *Clostridium difficile* occurred in two patients (<1%) in the antibiotics group and four (<1%) in the control group, and meticillin-resistant *Staphylococcus aureus* colonisation occurred in 11 patients (2%) in the antibiotics group and 14 (2%) in the control group.

Interpretation Antibiotic prophylaxis cannot be recommended for prevention of post-stroke pneumonia in patients [with] [dysphagia].

NO ANTIBIOTICS FOR PROPHYLAXIS

Background: Pn[eumonia] common acute complications after stroke and is ass[ociated] outcome. Biomarkers may help identifying stroke pat[ients] [develo]ping stroke-associated pneumonia (SAP) and to guide

trial investigated whether procalcitonin (PCT) ultrasensitive (PCTus)-guided [antibi]otic treatment of SAP can improve functional outcome after stroke.

Methods: In this international, multicenter, randomized, controlled clinical trial with blinded assessment of outcomes, patients with severe ischemic stroke in the middle cerebral artery territory were randomly assigned within 40 h after symptom onset to PCTus-based antibiotic therapy guidance in addition to stroke unit care or standard stroke unit care alone. The primary endpoint was functional outcome at 3 months, defined according to the modified Rankin Scale (mRS) and dichotomized as acceptable (≤4) or unacceptable (≥5). Secondary endpoints included usage of antibiotics, infection rates, days of fever, and mortality. The trial was registered with http://ClinicalTrials.gov (Identifier NCT01264549).

Results: In the intention-to-treat-analysis based on 227 patients (112 in PCT and 115 in control group), 197 patients completed the 3-month follow-up. Adherence to PCT guidance was 65%. PCT-guided therapy did not improve functional outcome as measured by mRS (odds ratio 0.79; 95% confidence interval 0.45–1.35, p = 0.47). Pneumonia rate and mortality were similar in both groups. Days with fever tended to be lower (p = 0.055), whereas total number of days treated with antibiotics were higher

The Randomized Controlled STRAWINSKI Trial: Procalcitonin-Guided Antibiotic Therapy after Stroke

(p = 0.004) in PCT compared to control group. A *post hoc* analysis including all PCT values in the intention-to-treat population demonstrated a significant increase on the first day of infection in patients with pneumonia and sepsis compared to patients with urinary tract infections or without infections ($p < 0.0001$).

Conclusion: PCTus-guided antibiotic therapy did not improve functional outcome at 3 months after severe ischemic stroke. PCT is a promising biomarker for early detection of pneumonia and sepsis in acute stroke patients.

Pressure sore Prevention

Think 'SSKIN' and help us to help you prevent pressure ulcers

S **Skin**

You should check your skin regularly.

S **Surface**

The surface you sit and lie on should be a good one; you may need a special mattress and possibly a special chair cushion.

K **Keep moving**

You should try and keep moving every two to four hours, even during the night. This is really important whether you are in bed, in a chair or in a wheelchair.

I **Incontinence**

Wet or soiled skin, due to incontinence, is weaker so please let someone know if you need help to wash your skin and change into clean clothes.

N **Nutrition**

Eating and drinking is really important so please try to eat even if you don't feel hungry. You may be offered a high energy drink or pudding to supplement your diet.

Pressure sore treatment by NPUAP

Stage	Treatment
Stage I	Wound protection with transparent film, preventive measures
Stage II	Dressings to maintain a moist wound environment
Stage III	Debridement of necrotic tissue, coverage with appropriate dressings, treatment of infection if present
Stage IV	Debridement of necrotic tissue, coverage with appropriate dressings, treatment of infection if present, surgery if necessary

REFERENCES

- Harrison's principles of internal medicine 20th edition
- J. P. Mohr, Philip A. Wolf, et al. | 13 May 2011,stroke:pathophysiology,diagnosis and management
- Stroke. 2019;50:773-778
- Neuroanatomy through Clinical Cases.2010.2nd ed - Hal Blumenfeld

THANK YOU